

EE-102: Electric Circuit Analysis

LECTURE NO 6:

Network Theorems

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Organization of Lecture

1. Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Maximum Power Transfer Theorem

1. Superposition Theorem (1/15)

- The *superposition theorem* is unquestionably one of the most powerful in this field. It has such widespread application that people often apply it without recognizing that their manoeuvres are valid only because of this theorem. In general, the theorem can be used to do the following:
 - *Analyse networks such as that have two or more sources that are not in series or parallel.*
 - *Reveal the effect of each source on a particular quantity of interest.*
 - *For sources of different types (such as dc and ac which affect the parameters of the network in a different manner), apply a separate analysis for each type, with the total result.*

1. Superposition Theorem (2/15)

- *The current through, or voltage across, any element of a network is equal to the algebraic sum of the currents or voltages produced independently by each source.*
- *When removing a voltage source from a network schematic, replace it with a direct connection (short circuit) of zero ohms. Any internal resistance associated with the source must remain in the network.*
- *When removing a current source from a network schematic, replace it by an open circuit of infinite ohms. Any internal resistance associated with the source must remain in the network.*

1. Superposition Theorem (3/15)

- In other words, this theorem allows us to find a solution for a current or voltage using only one source at a time. Once we have the solution for each source, we can combine the results to obtain the total solution. If we are to consider the effects of each source, the other sources obviously must be removed. The above statements are illustrated in Fig.1.

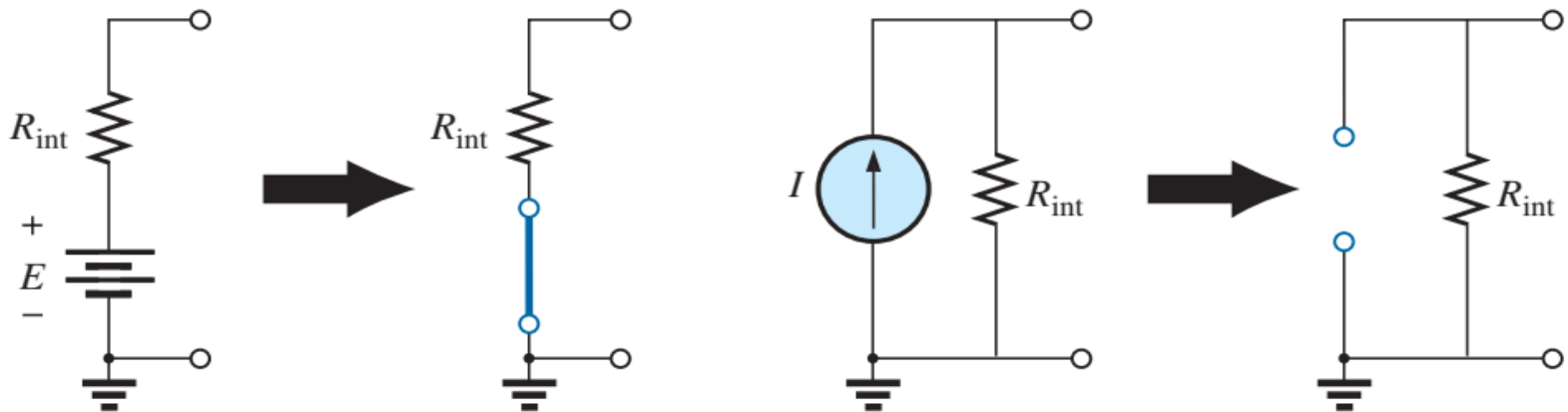


FIG. 1

Removing a voltage source and a current source to permit the application of the superposition theorem.

1. Superposition Theorem (4/15)

- *Superposition cannot be applied to power effects* because the power is related to the square of the voltage across a resistor or the current through a resistor. The squared term results in a nonlinear (a curve, not a straight line) relationship between the power and the determining current or voltage. For example, doubling the current through a resistor does not double the power to the resistor (as defined by a linear relationship) but, in fact, increases it by a factor of 4 (due to the squared term). Tripling the current increases the power level by a factor of 9.

1. Superposition Theorem (5/15)

EXAMPLE 1

- Using the superposition theorem, determine the current through resistor R_2 for the network in Fig. 2.
- Demonstrate that the superposition theorem is not applicable to power levels.

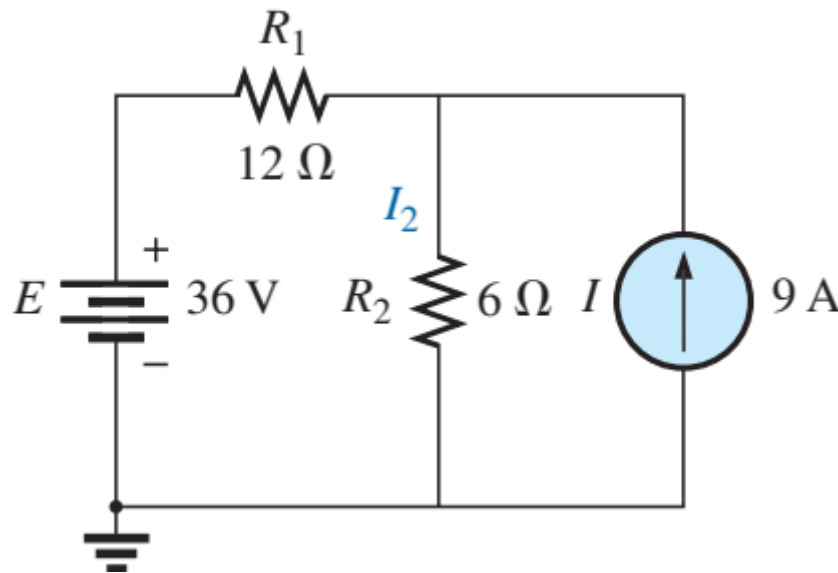


FIG. 2

1. Superposition Theorem (6/15)

Solutions:

- a. In order to determine the effect of the 36 V voltage source, the current source must be replaced by an open-circuit equivalent as shown in Fig. 3. The result is a simple series circuit with a current equal to

$$I'_2 = \frac{E}{R_T} = \frac{E}{R_1 + R_2} = \frac{36 \text{ V}}{12 \Omega + 6 \Omega} = \frac{36 \text{ V}}{18 \Omega} = 2 \text{ A}$$

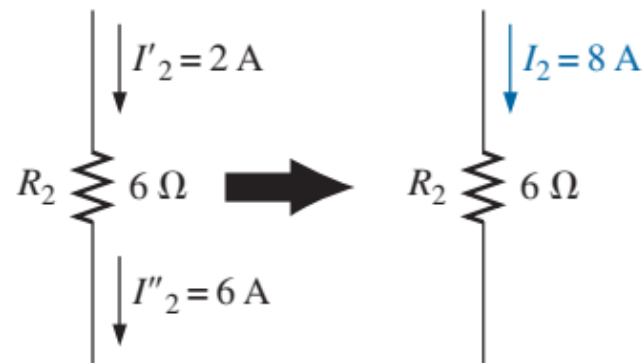
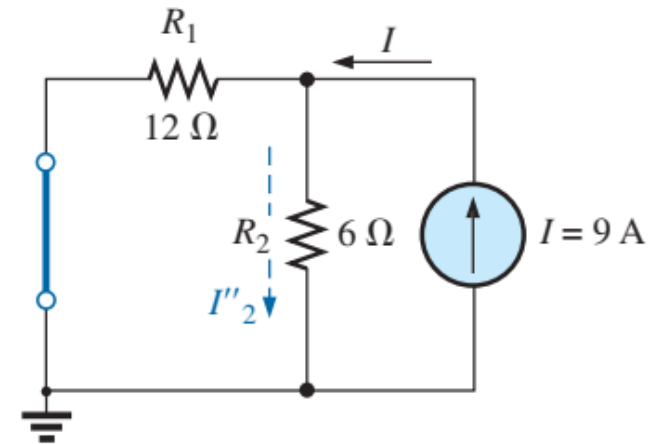
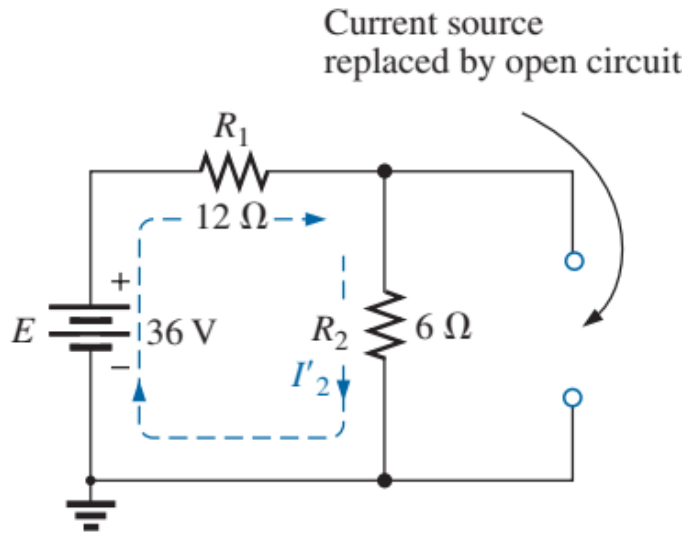
Examining the effect of the 9 A current source requires replacing the 36 V voltage source by a short-circuit equivalent as shown in Fig. 4. The result is a parallel combination of resistors R_1 and R_2 . Applying the current divider rule results in

$$I''_2 = \frac{R_1(I)}{R_1 + R_2} = \frac{(12 \Omega)(9 \text{ A})}{12 \Omega + 6 \Omega} = 6 \text{ A}$$

Since the contribution to current I_2 has the same direction for each source, as shown in Fig. 5, the total solution for current I_2 is the sum of the currents established by the two sources. That is,

$$I_2 = I'_2 + I''_2 = 2 \text{ A} + 6 \text{ A} = \mathbf{8 \text{ A}}$$

1. Superposition Theorem (7/15)



1. Superposition Theorem (8/15)

- b. Using Fig. 3 and the results obtained, we find the power delivered to the $6\ \Omega$ resistor

$$P_1 = (I'_2)^2(R_2) = (2\ \text{A})^2(6\ \Omega) = \mathbf{24\ \text{W}}$$

Using Fig. 4 and the results obtained, we find the power delivered to the $6\ \Omega$ resistor

$$P_2 = (I''_2)^2(R_2) = (6\ \text{A})^2(6\ \Omega) = \mathbf{216\ \text{W}}$$

Using the total results of Fig. 5, we obtain the power delivered to the $6\ \Omega$ resistor

$$P_T = I_2^2 R_2 = (8\ \text{A})^2(6\ \Omega) = \mathbf{384\ \text{W}}$$

It is now quite clear that the power delivered to the $6\ \Omega$ resistor using the total current of $8\ \text{A}$ is not equal to the sum of the power levels due to each source independently. That is,

$$P_1 + P_2 = 24\ \text{W} + 216\ \text{W} = 240\ \text{W} \neq P_T = 384\ \text{W}$$

1. Superposition Theorem (9/15)

Homework!

EXAMPLE 2 Using the superposition theorem, determine the current through the $12\ \Omega$ resistor in Fig. 6. Note that this is a two-source network of the type examined when we applied branch-current analysis and mesh analysis.

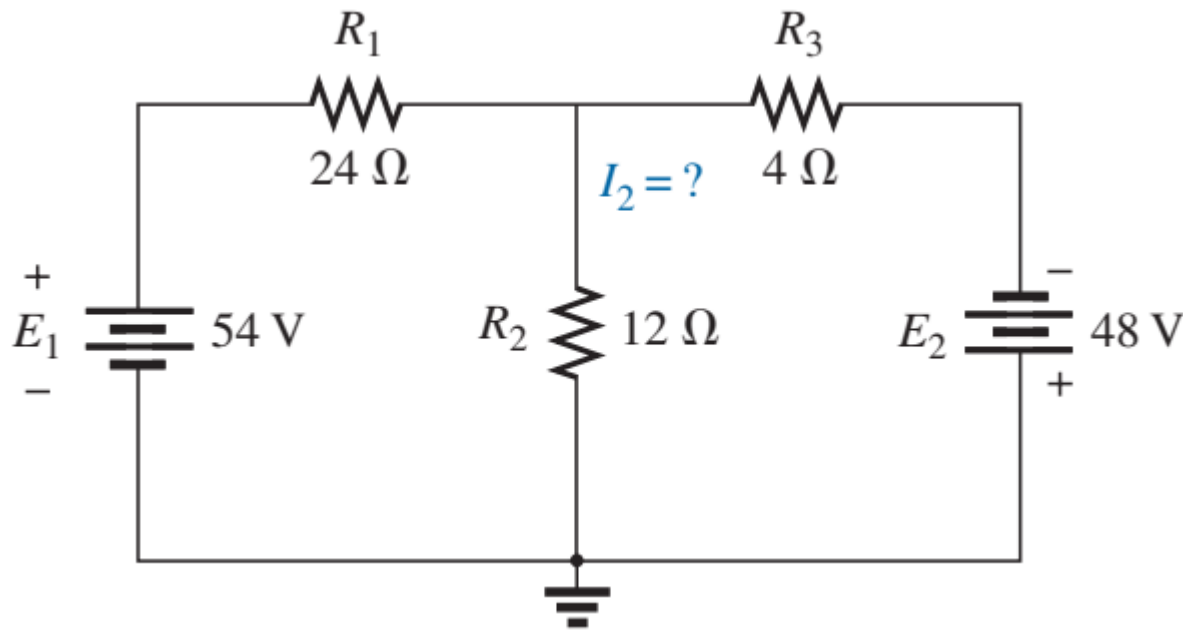


Fig. 6.

1. Superposition Theorem (10/15)

Homework!

EXAMPLE 3 Using the superposition theorem, determine current I_1 for the network in Fig. 7.

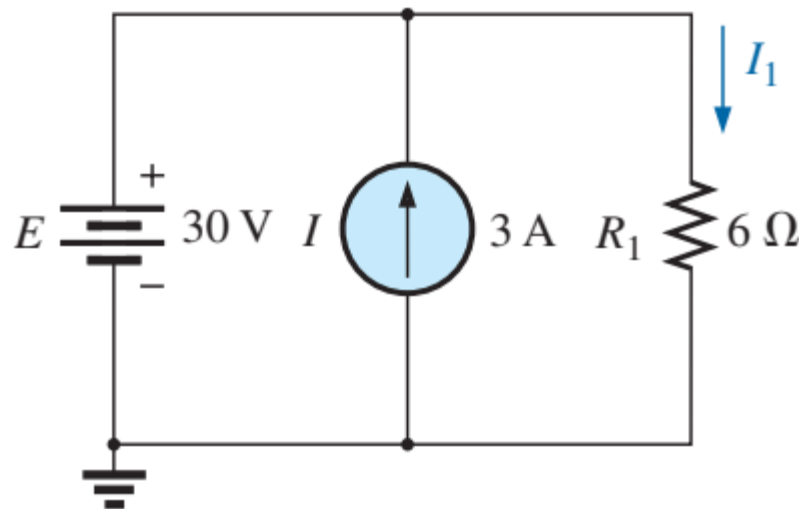


Fig. 7.

1. Superposition Theorem (11/15)

Homework!

EXAMPLE 4 Using the principle of superposition, find the current I_2 through the $12\text{ k}\Omega$ resistor in Fig.8.

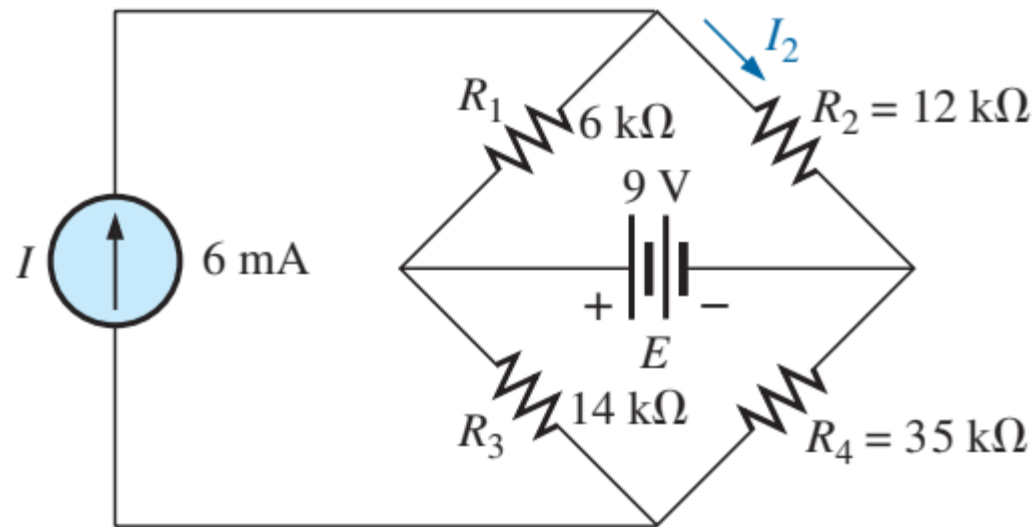


Fig. 8.

1. Superposition Theorem (12/15)

EXAMPLE 5 Find the current through the $2\ \Omega$ resistor of the network in Fig. 9. The presence of three sources results in three different networks to be analyzed.

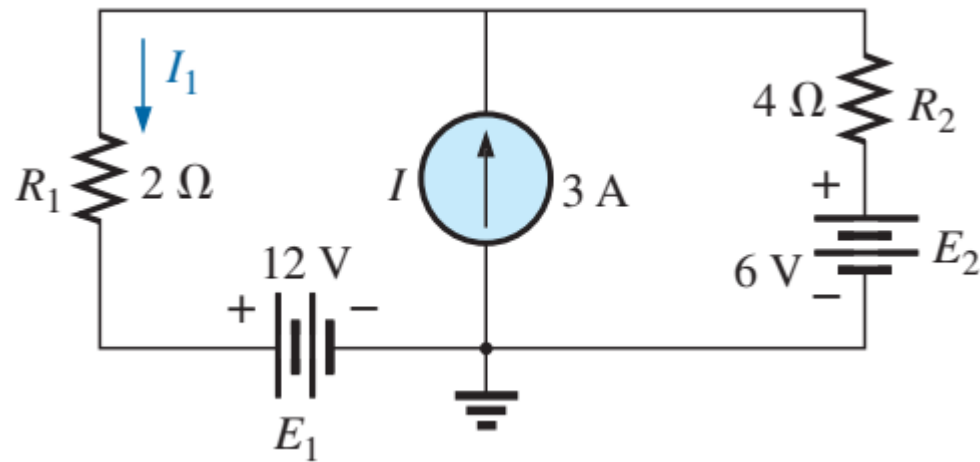


Fig. 9.

1. Superposition Theorem (13/15)

Solution: Consider the effect of the 12 V source (Fig.10.):

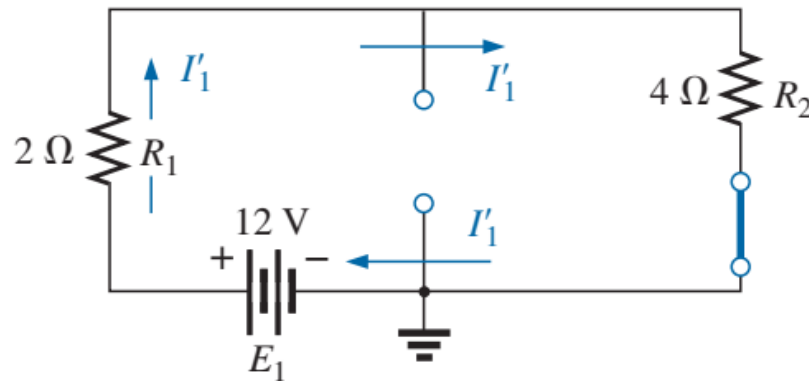


Fig. 10.

$$I'_1 = \frac{E_1}{R_1 + R_2} = \frac{12\text{ V}}{2\Omega + 4\Omega} = \frac{12\text{ V}}{6\Omega} = 2\text{ A}$$

Consider the effect of the 6 V source (Fig.11.):

$$I''_1 = \frac{E_2}{R_1 + R_2} = \frac{6\text{ V}}{2\Omega + 4\Omega} = \frac{6\text{ V}}{6\Omega} = 1\text{ A}$$

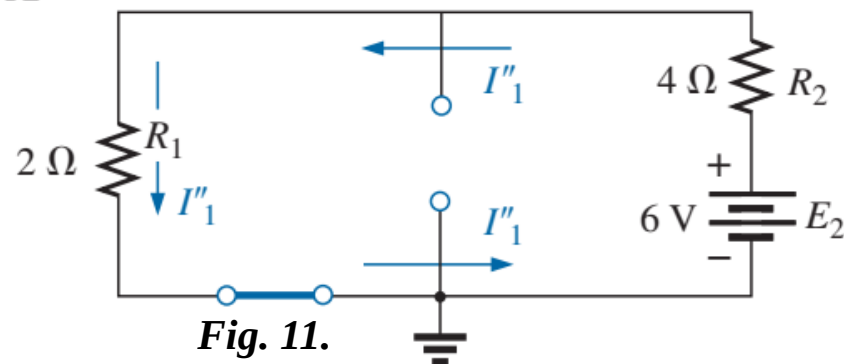
1. Superposition Theorem (14/15)

Consider the effect of the 3 A source (Fig.12.): Applying the current divider rule gives

$$I'''_1 = \frac{R_2 I}{R_1 + R_2} = \frac{(4\ \Omega)(3\ \text{A})}{2\ \Omega + 4\ \Omega} = \frac{12\ \text{A}}{6} = 2\ \text{A}$$

The total current through the $2\ \Omega$ resistor appears in Fig.13, and

$$\begin{aligned} I_1 &= I''_1 + I'''_1 - I'_1 \\ &= 1\ \text{A} + 2\ \text{A} - 2\ \text{A} = 1\ \text{A} \end{aligned}$$



1. Superposition Theorem (15/15)

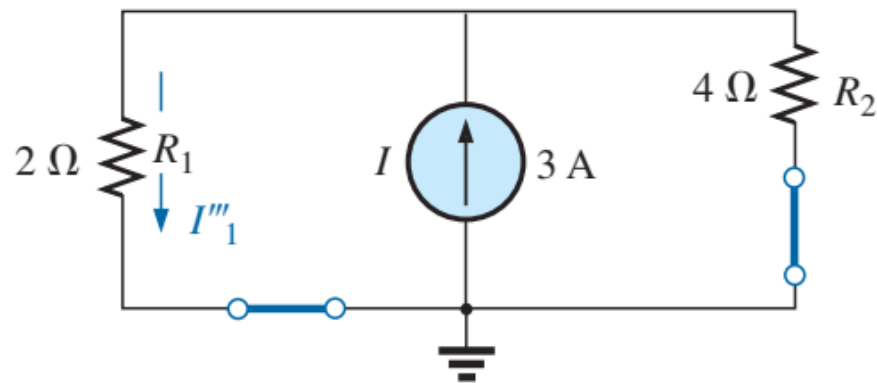


Fig. 12.

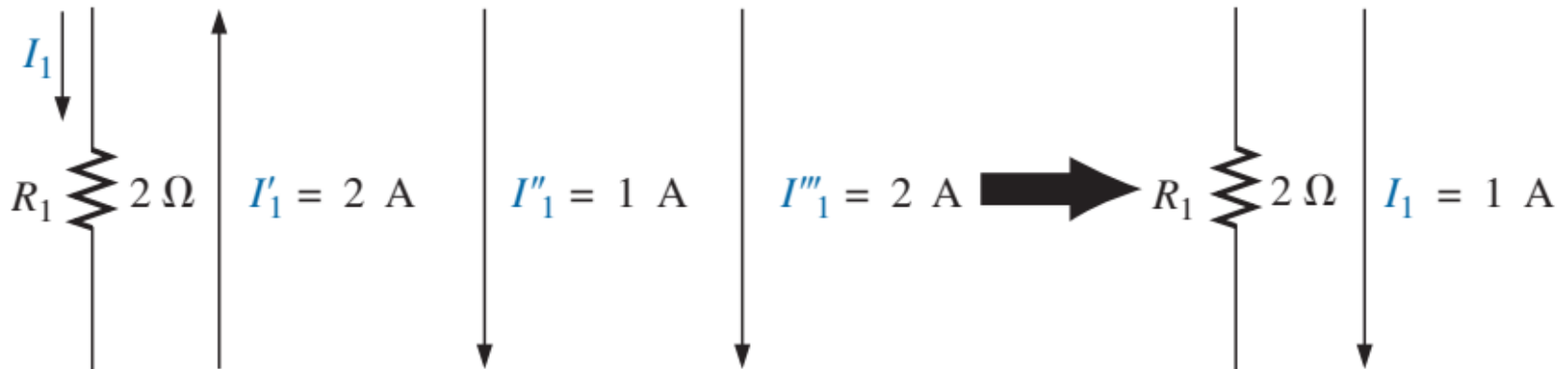


Fig. 13.

2. Thévenin's Theorem (1/19)

- **Thévenin's theorem** allows even the most complicated circuit to be reduced to a single voltage source and a single resistance.
- The following steps provide a technique that converts any complex circuit into its Thévenin equivalent:
 1. *Identify and remove the load from the circuit.*
 2. *Label the resulting two terminals. We will label them as **a** and **b**, although any notation may be used.*
 3. *Set all sources in the circuit to zero. Voltage sources are set to zero by replacing them with short circuits (zero volts). Current sources are set to zero by replacing them with open circuits (zero amps).*
 4. *Determine the Thévenin equivalent resistance, R_{Th} , by calculating the resistance “seen” between terminals a and b. It may be necessary to redraw the circuit to simplify this step.*

2. Thévenin's Theorem (2/19)

5. Replace the sources removed in Step 3, and determine the open circuit voltage between the terminals. *If the circuit has more than one source, it may be necessary to use the superposition theorem.* In that case, it will be necessary to determine the open circuit voltage due to each source separately and then determine the combined effect. The resulting open circuit voltage will be the value of the Thévenin voltage, E_{Th} .
6. Draw the Thévenin equivalent circuit using the resistance determined in Step 4 and the voltage calculated in Step 5. As part of the resulting circuit, include that portion of the network removed in Step 1.

2. Thévenin's Theorem (3/19)

- Any two-terminal dc network can be replaced by an equivalent circuit consisting solely of a voltage source and a series resistor as shown in Fig. 14.

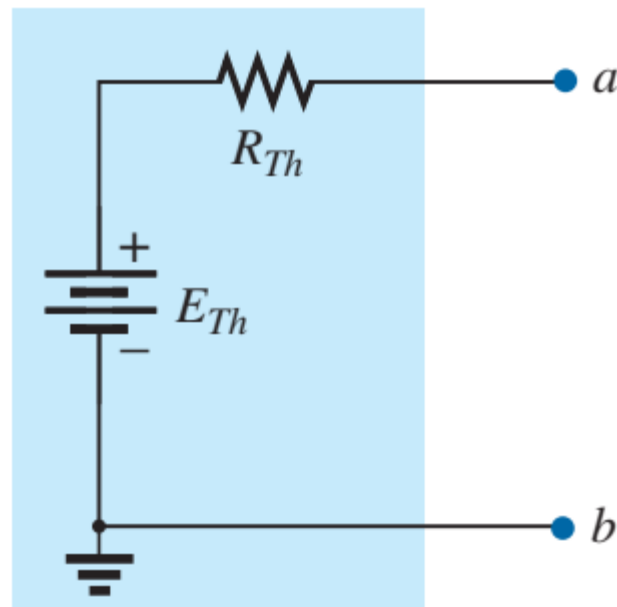


Fig. 14: Thévenin equivalent circuit

2. Thévenin's Theorem (4/19)

EXAMPLE 6 Find the Thévenin equivalent circuit for the network in the shaded area of the network in Fig. 15. Then find the current through R_L for values of $2\ \Omega$, $10\ \Omega$, and $100\ \Omega$.

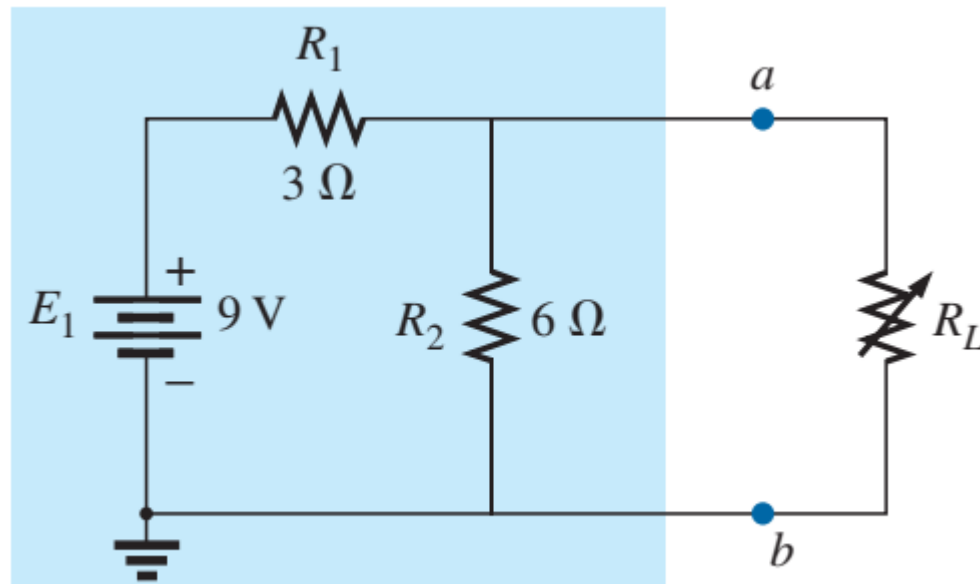
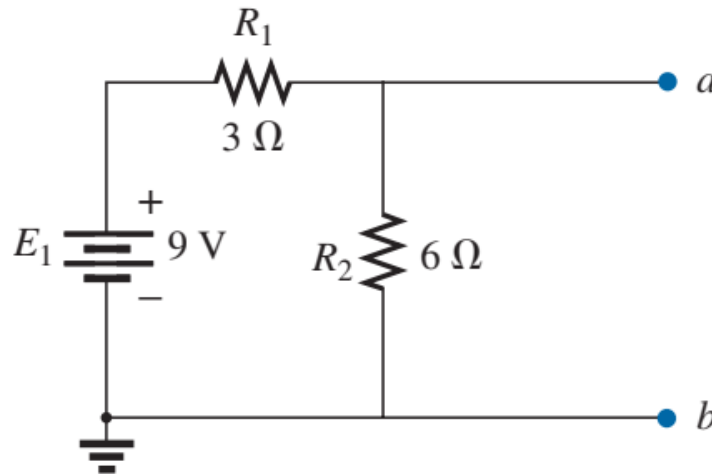


Fig. 15.

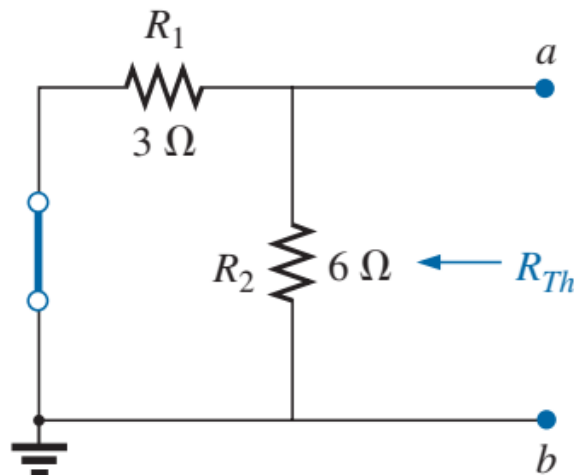
2. Thévenin's Theorem (5/19)

Solution:

Step 1&2:



Step 3&4:

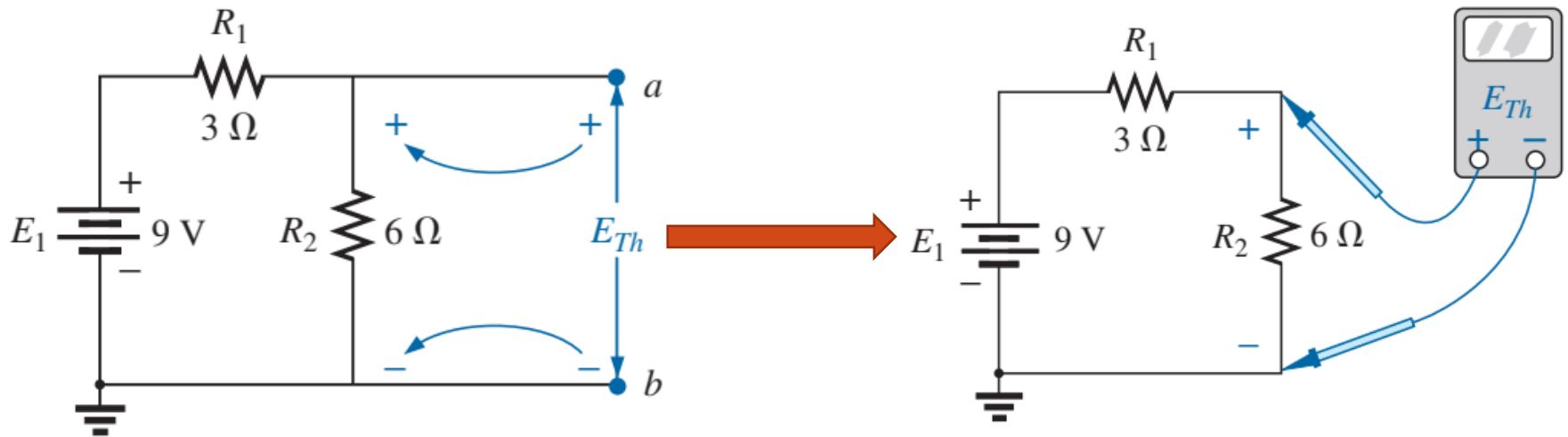


$$R_{Th} = R_1 \parallel R_2 = \frac{(3\ \Omega)(6\ \Omega)}{3\ \Omega + 6\ \Omega} = \mathbf{2\ \Omega}$$

2. Thévenin's Theorem (6/19)

Solution:

Step 5: Replace the voltage source. For this case, the open circuit voltage E_{Th} is the same as the voltage drop across the 6Ω resistor. Applying the voltage divider rule gives

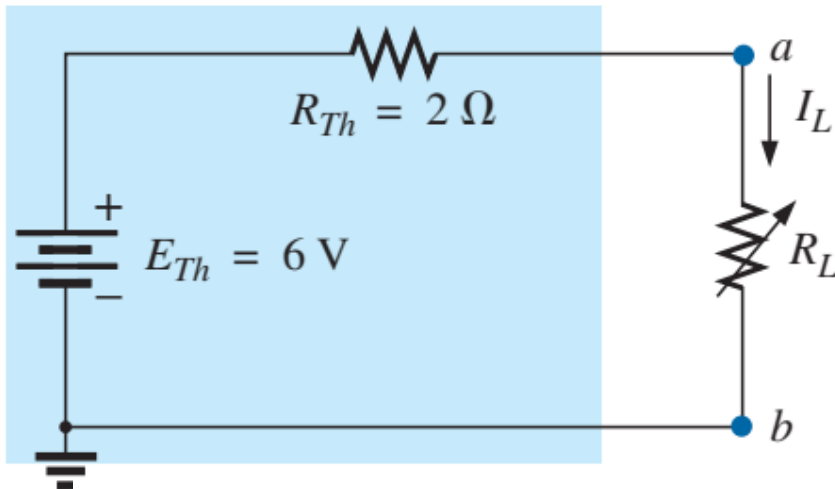


$$E_{Th} = \frac{R_2 E_1}{R_2 + R_1} = \frac{(6\Omega)(9\text{ V})}{6\Omega + 3\Omega} = \frac{54\text{ V}}{9} = 6\text{ V}$$

2. Thévenin's Theorem (7/19)

Solution:

Step 6:



$$I_L = \frac{E_{Th}}{R_{Th} + R_L}$$

$$R_L = 2 \Omega:$$

$$I_L = \frac{6 \text{ V}}{2 \Omega + 2 \Omega} = \mathbf{1.5 \text{ A}}$$

$$R_L = 10 \Omega:$$

$$I_L = \frac{6 \text{ V}}{2 \Omega + 10 \Omega} = \mathbf{0.5 \text{ A}}$$

$$R_L = 100 \Omega:$$

$$I_L = \frac{6 \text{ V}}{2 \Omega + 100 \Omega} = \mathbf{0.06 \text{ A}}$$

2. Thévenin's Theorem (8/19)

Homework!

EXAMPLE 7 Find the Thévenin equivalent circuit for the network in the shaded area of the network in Fig. 16.

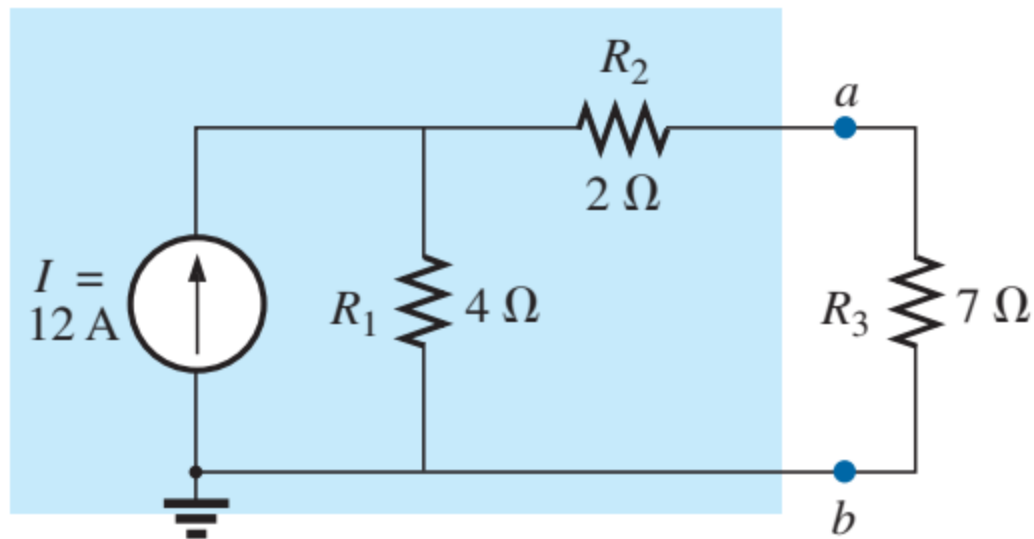


Fig. 16.

2. Thévenin's Theorem (9/19)

EXAMPLE 8 Find the Thévenin equivalent circuit for the network in the shaded area of the network in Fig.17.

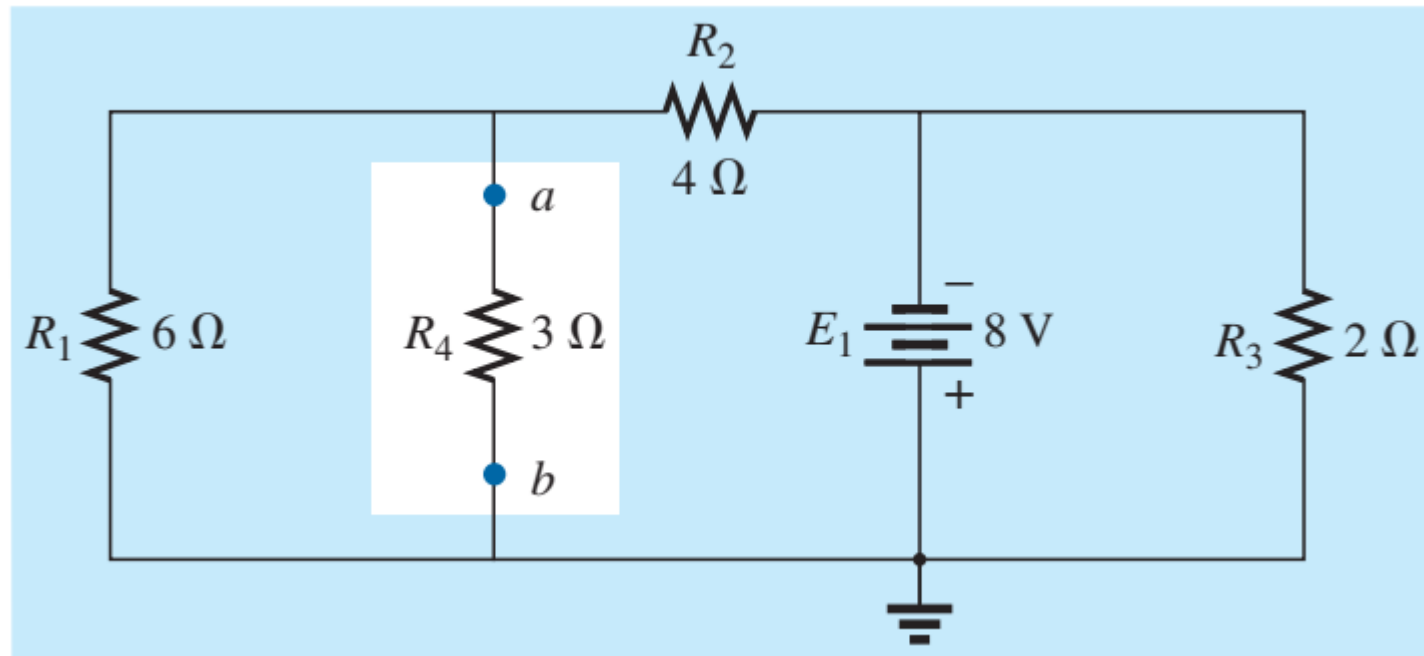
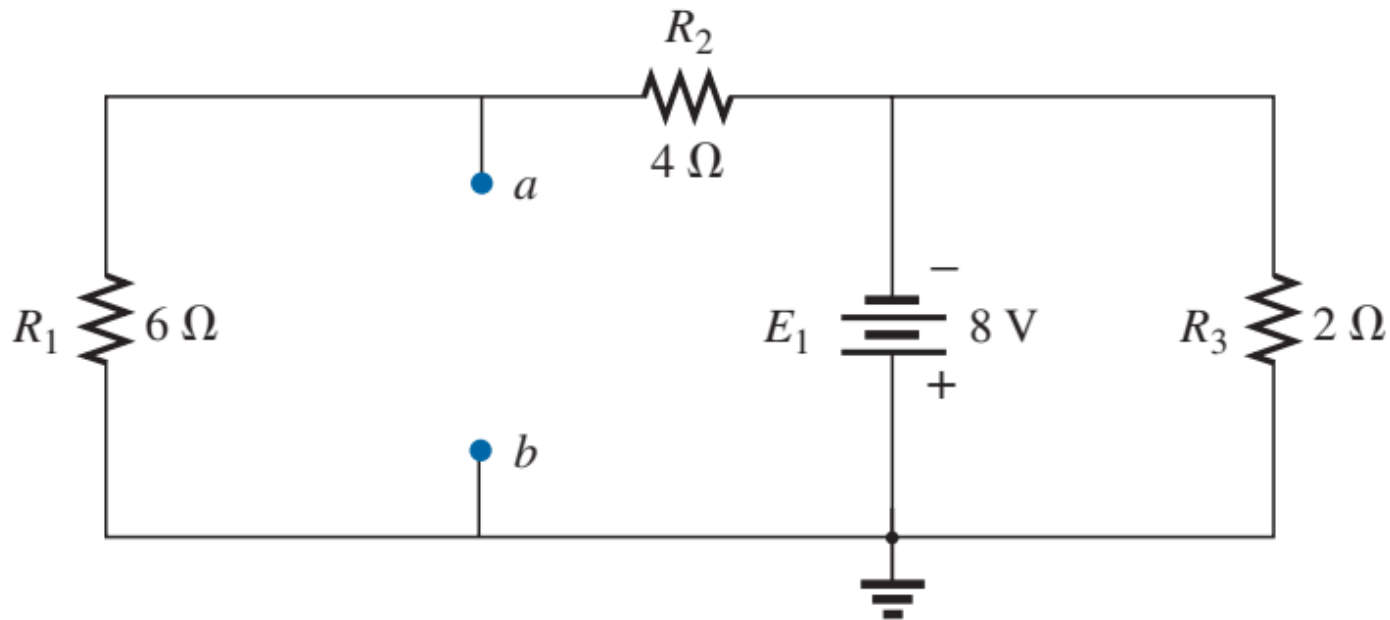


Fig. 17.

2. Thévenin's Theorem (10/19)

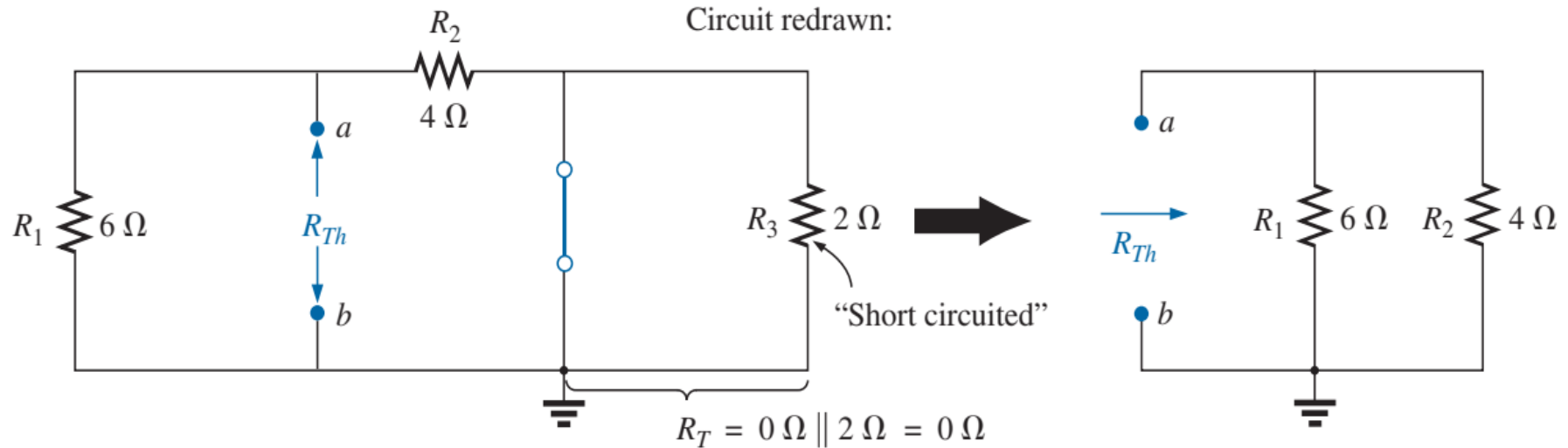
Solution:

Step 1&2:



2. Thévenin's Theorem (11/19)

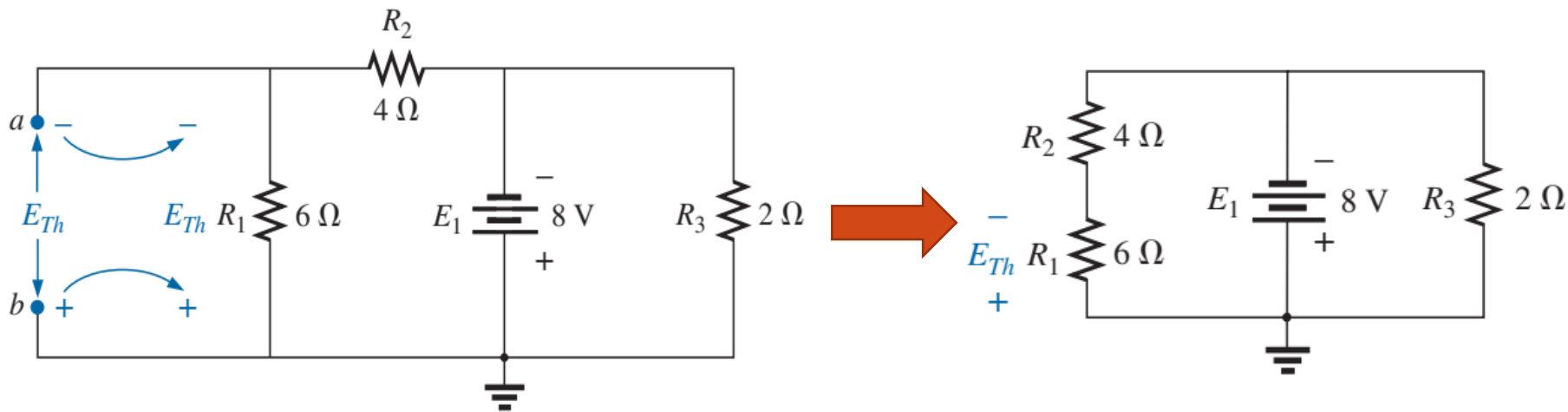
Step 3&4:



$$R_{Th} = R_1 \parallel R_2 = \frac{(6\ \Omega)(4\ \Omega)}{6\ \Omega + 4\ \Omega} = \frac{24\ \Omega}{10} = \mathbf{2.4\ \Omega}$$

2. Thévenin's Theorem (12/19)

Step 5:

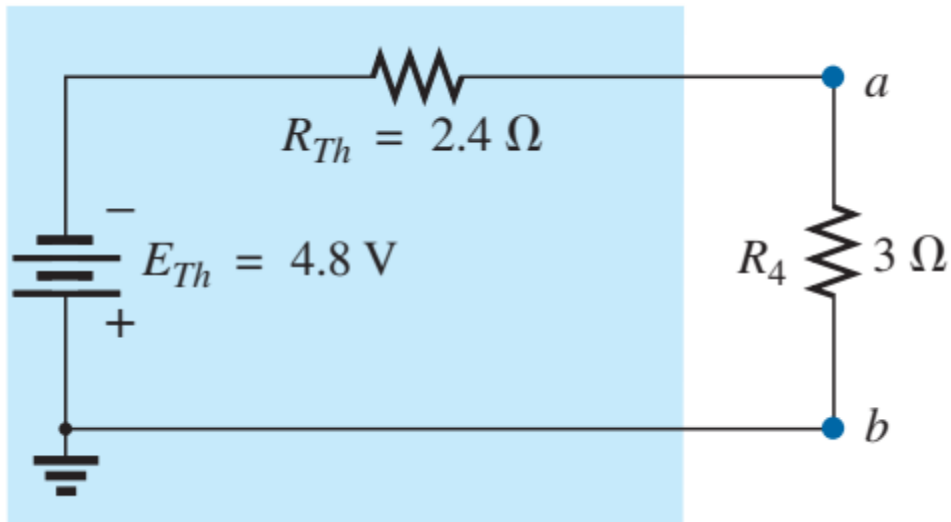


- Applying the voltage divider rule gives:

$$E_{Th} = \frac{R_1 E_1}{R_1 + R_2} = \frac{(6\ \Omega)(8\text{ V})}{6\ \Omega + 4\ \Omega} = \frac{48\text{ V}}{10} = 4.8\text{ V}$$

2. Thévenin's Theorem (13/19)

Step 6:



2. Thévenin's Theorem (14/19)

EXAMPLE 9 (Two sources) Find the Thévenin circuit for the network within the shaded area of Fig. 18.

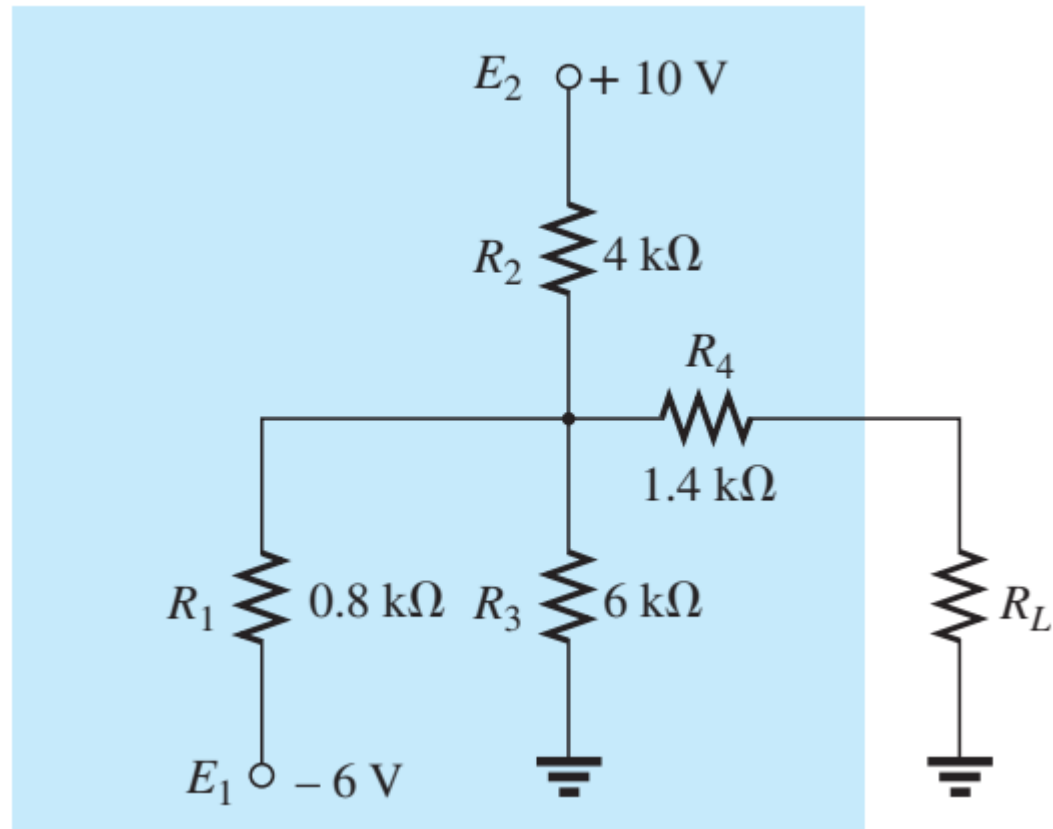
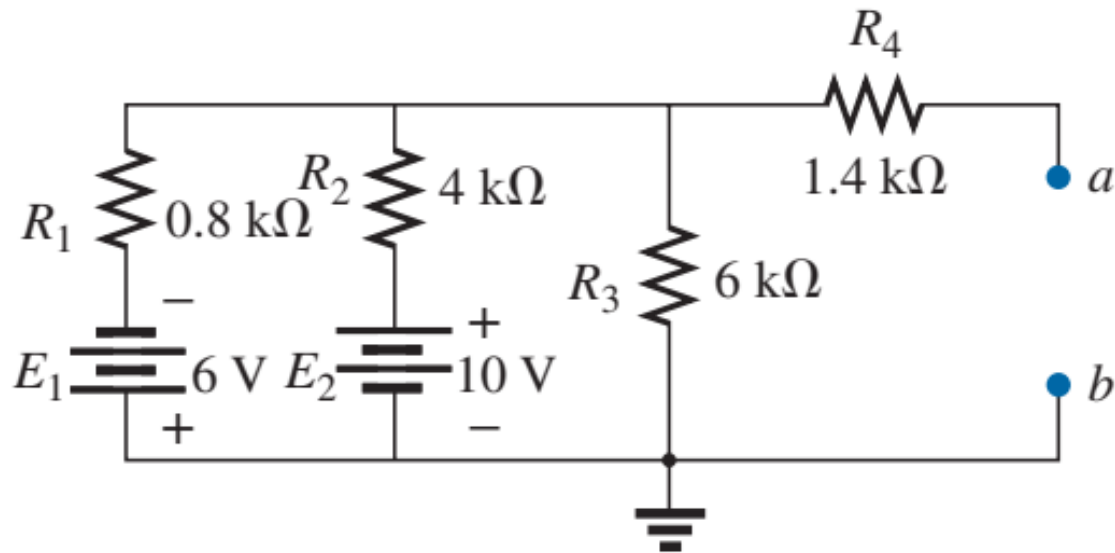


Fig. 18.

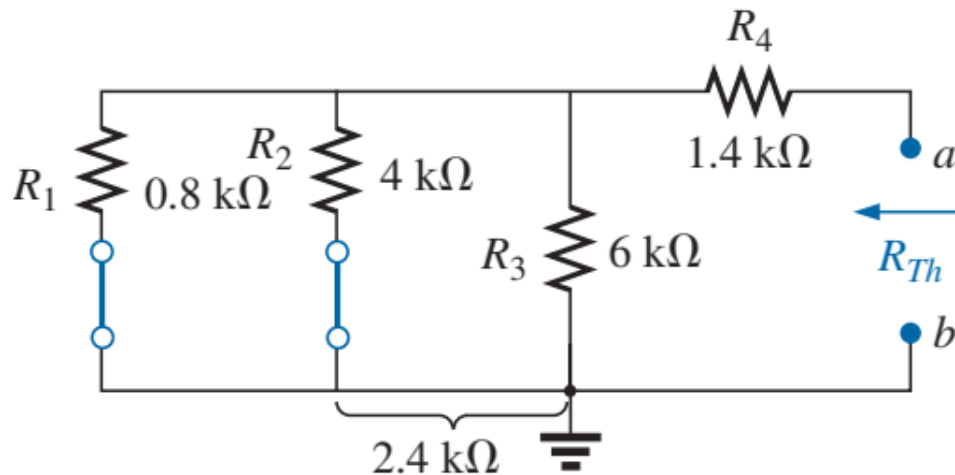
2. Thévenin's Theorem (15/19)

Step 1&2: Figure 18 can be redrawn as:



2. Thévenin's Theorem (16/19)

Step 3&4:



$$\begin{aligned} R_{Th} &= R_4 + R_1 \parallel R_2 \parallel R_3 \\ &= 1.4\text{ k}\Omega + 0.8\text{ k}\Omega \parallel 4\text{ k}\Omega \parallel 6\text{ k}\Omega \\ &= 1.4\text{ k}\Omega + 0.8\text{ k}\Omega \parallel 2.4\text{ k}\Omega \\ &= 1.4\text{ k}\Omega + 0.6\text{ k}\Omega \\ &= \mathbf{2\text{ k}\Omega} \end{aligned}$$

2. Thévenin's Theorem (17/19)

Step 5:

- Applying superposition, we will consider the effects of the voltage source E_1 first. Note Fig. below. The open circuit requires that $V_4 = I_4 R_4 = (0) R_4 = 0$ V, and

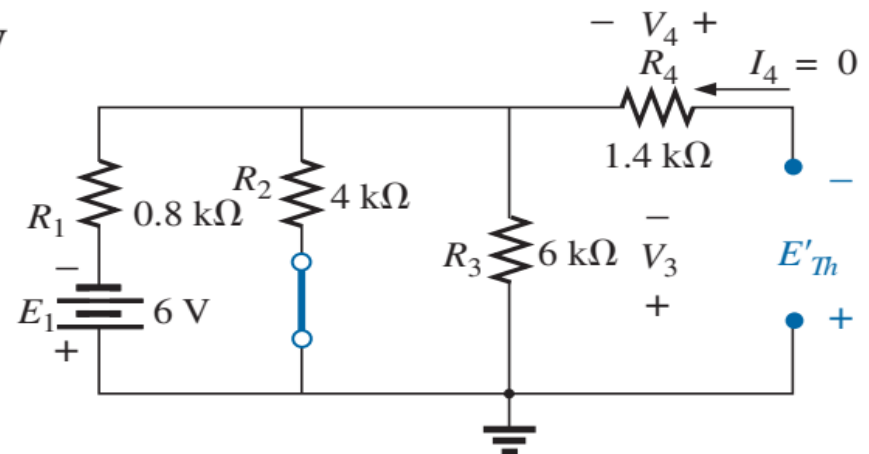
$$E'_{Th} = V_3$$

$$R'_T = R_2 \parallel R_3 = 4 \text{ k}\Omega \parallel 6 \text{ k}\Omega = 2.4 \text{ k}\Omega$$

Applying the voltage divider rule,

$$V_3 = \frac{R'_T E_1}{R'_T + R_1} = \frac{(2.4 \text{ k}\Omega)(6 \text{ V})}{2.4 \text{ k}\Omega + 0.8 \text{ k}\Omega} = \frac{14.4 \text{ V}}{3.2} = 4.5 \text{ V}$$

$$E'_{Th} = V_3 = 4.5 \text{ V}$$



2. Thévenin's Theorem (18/19)

Step 5: (continued)

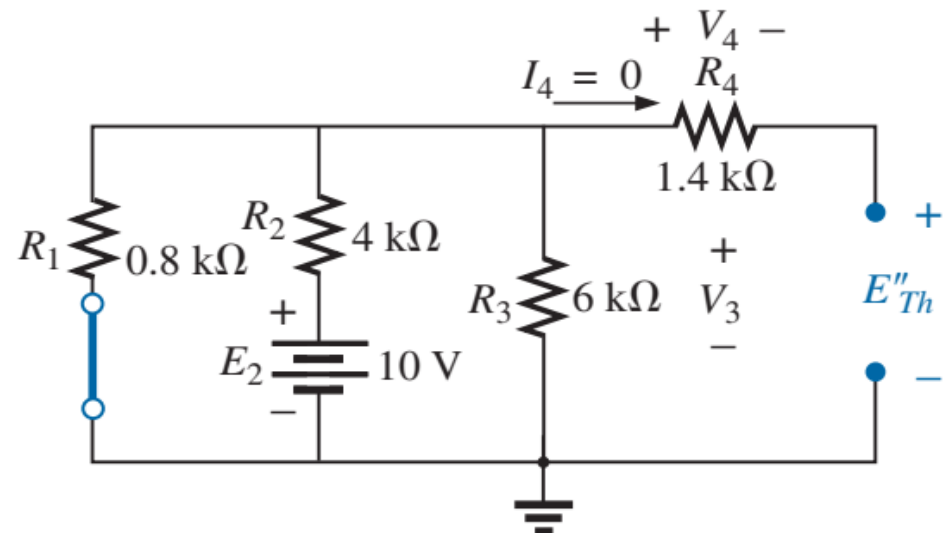
- For the voltage source E_2 , the network in Fig. below results. Again, $V_4 = I_4 R_4 = (0)R_4 = 0$ V, and

$$E''_{Th} = V_3$$

$$R'_T = R_1 \parallel R_3 = 0.8 \text{ k}\Omega \parallel 6 \text{ k}\Omega = 0.706 \text{ k}\Omega$$

$$V_3 = \frac{R'_T E_2}{R'_T + R_2} = \frac{(0.706 \text{ k}\Omega)(10 \text{ V})}{0.706 \text{ k}\Omega + 4 \text{ k}\Omega} = \frac{7.06 \text{ V}}{4.706} = 1.5 \text{ V}$$

$$E''_{Th} = V_3 = 1.5 \text{ V}$$

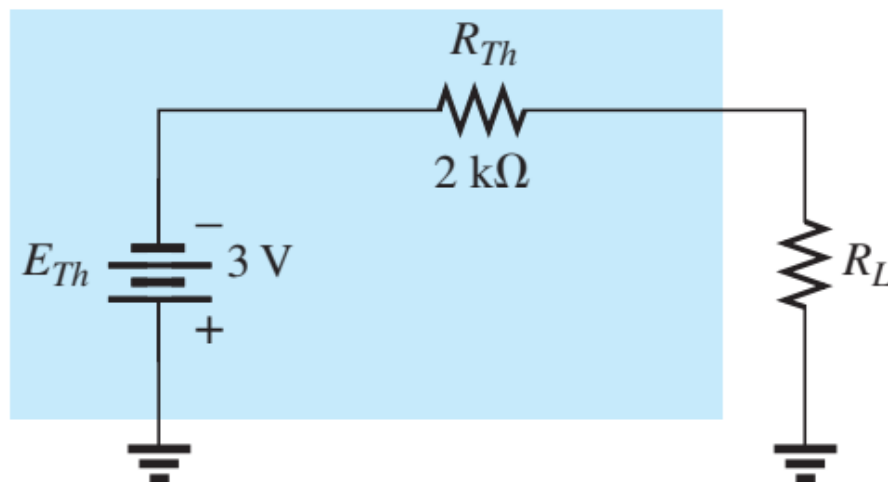


2. Thévenin's Theorem (19/19)

Step 6:

Since E'_{Th} and E''_{Th} have opposite polarities,

$$\begin{aligned} E_{Th} &= E'_{Th} - E''_{Th} \\ &= 4.5 \text{ V} - 1.5 \text{ V} \\ &= \mathbf{3 \text{ V}} \quad (\text{polarity of } E'_{Th}) \end{aligned}$$



3. Norton's Theorem (1/13)

- **Norton's theorem** is a circuit analysis technique that is similar to Thévenin's theorem. By using this theorem we reduce the circuit to a single current source and one parallel resistor.
- The following steps provide a technique that allows the conversion of any circuit into its Norton equivalent:
 1. *Identify and remove the load from the circuit.*
 2. *Label the resulting two terminals. We will label them as **a** and **b**, although any notation may be used.*
 3. *Set all sources to zero. As before, voltage sources are set to zero by replacing them with short circuits, and current sources are set to zero by replacing them with open circuits.*
 4. *Determine the Norton equivalent resistance, $\mathbf{R_N}$, by calculating the resistance seen between terminals a and b. It may be necessary to redraw the circuit to simplify this step.*

3. Norton's Theorem (2/13)

5. Replace the sources removed in Step 3, and *determine the current that would occur in a short if the short were connected between terminals a and b. If the original circuit has more than one source, it may be necessary to use the superposition theorem.* In this case, it will be necessary to determine the short-circuit current due to each source separately and then determine the combined effect. The resulting short-circuit current will be the value of the Norton current I_N .
6. Sketch the Norton equivalent circuit using the resistance determined in Step 4 and the current calculated in Step 5. As part of the resulting circuit, include that portion of the network removed in Step 1.

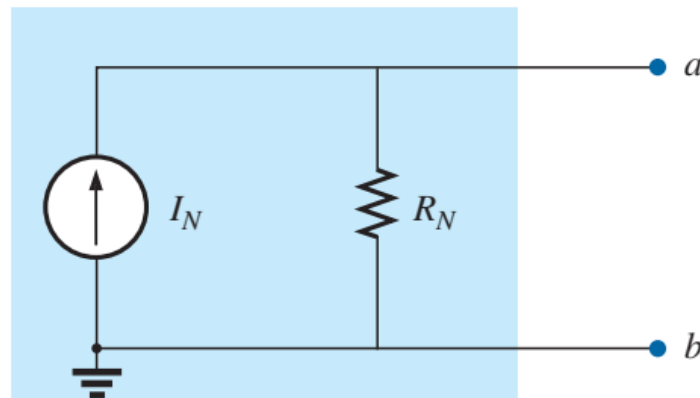


Fig. 19: Norton equivalent circuit

3. Norton's Theorem (3/13)

- The Norton equivalent circuit may also be determined directly from the Thévenin equivalent circuit by using the source conversion technique developed earlier.

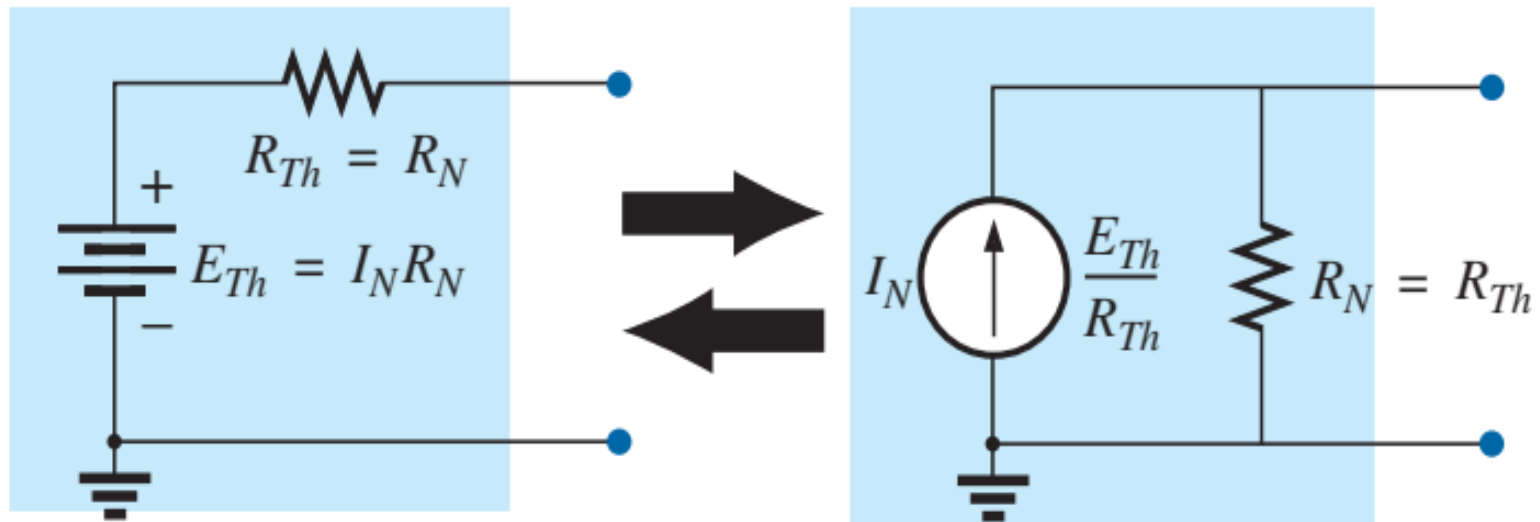


Fig. 20: Converting between Thévenin and Norton equivalent circuits.

3. Norton's Theorem (4/13)

EXAMPLE 10 Find the Norton equivalent circuit for the network in the shaded area in Fig. 21.

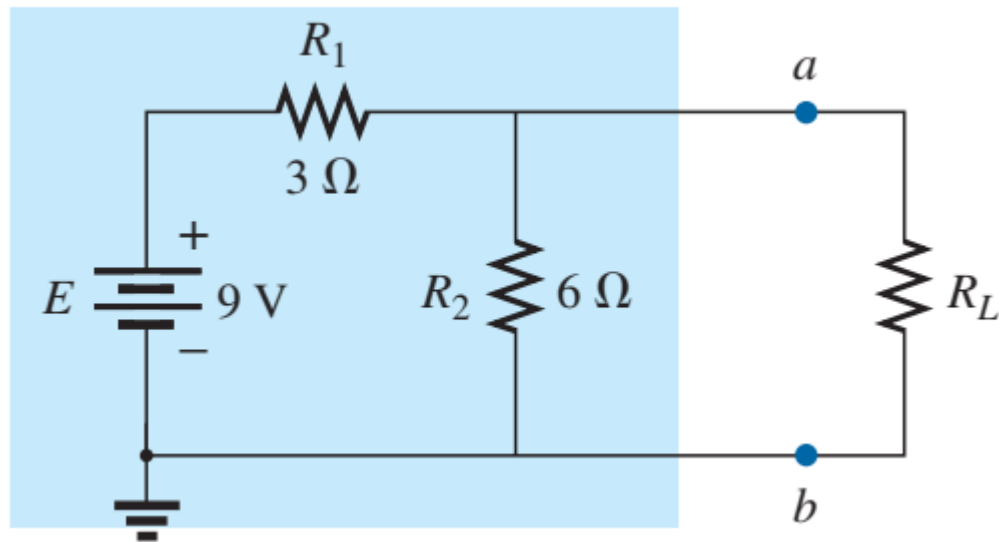
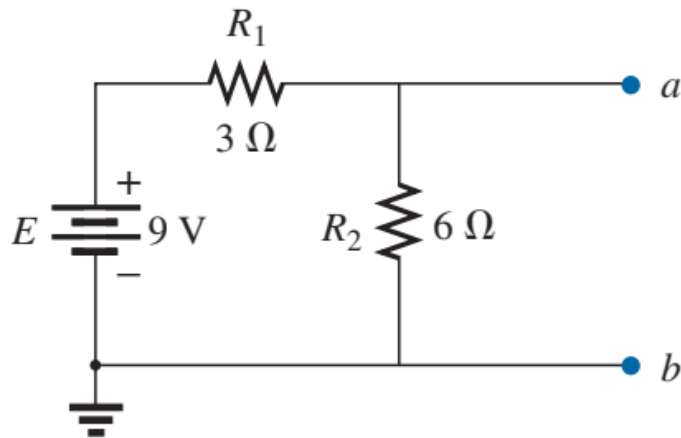


Fig. 21.

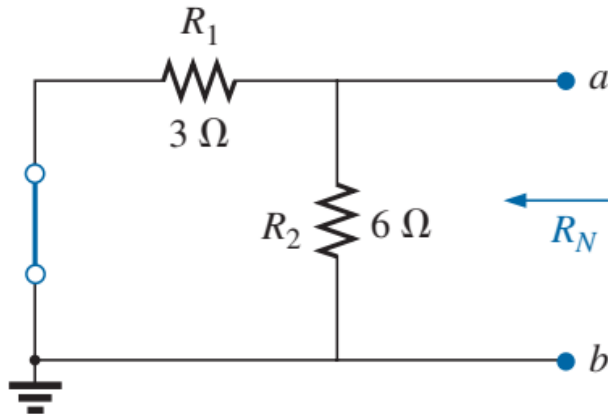
3. Norton's Theorem (5/13)

Solution:

Step 1&2:



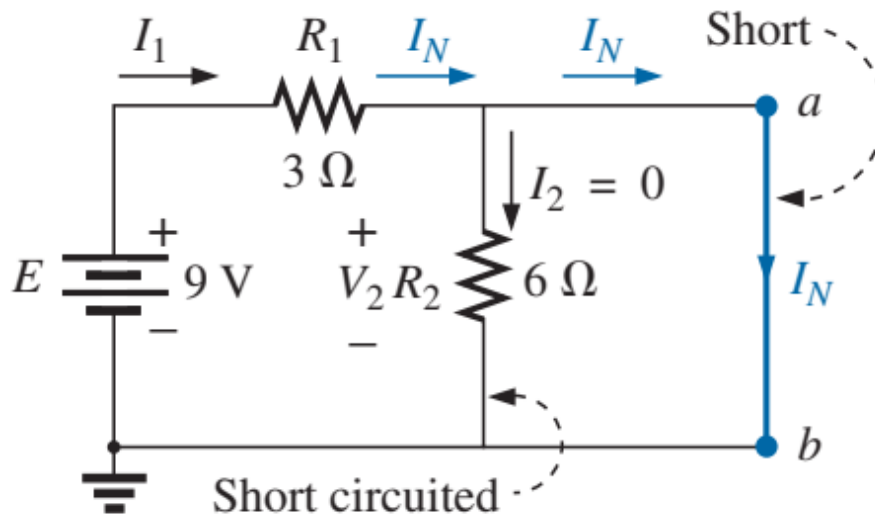
Step 3&4:



$$R_N = R_1 \parallel R_2 = 3\Omega \parallel 6\Omega = \frac{(3\Omega)(6\Omega)}{3\Omega + 6\Omega} = \frac{18\Omega}{9} = 2\Omega$$

3. Norton's Theorem (6/13)

Step 5:



$$V_2 = I_2 R_2 = (0)6\Omega = 0\text{ V}$$

Therefore,

$$I_N = \frac{E}{R_1} = \frac{9\text{ V}}{3\Omega} = 3\text{ A}$$

Remember!

$$I_1 = I_2 + I_N$$

$$I_1 = 0 + I_N$$

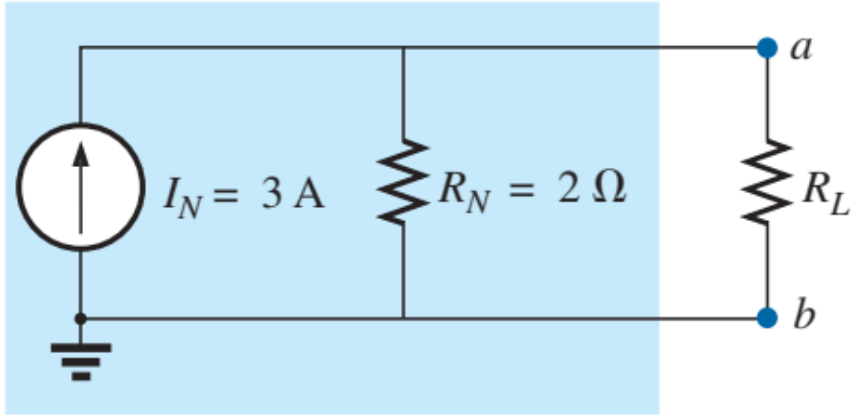
$$I_1 = I_N$$

$$I_1 = E/R_1$$

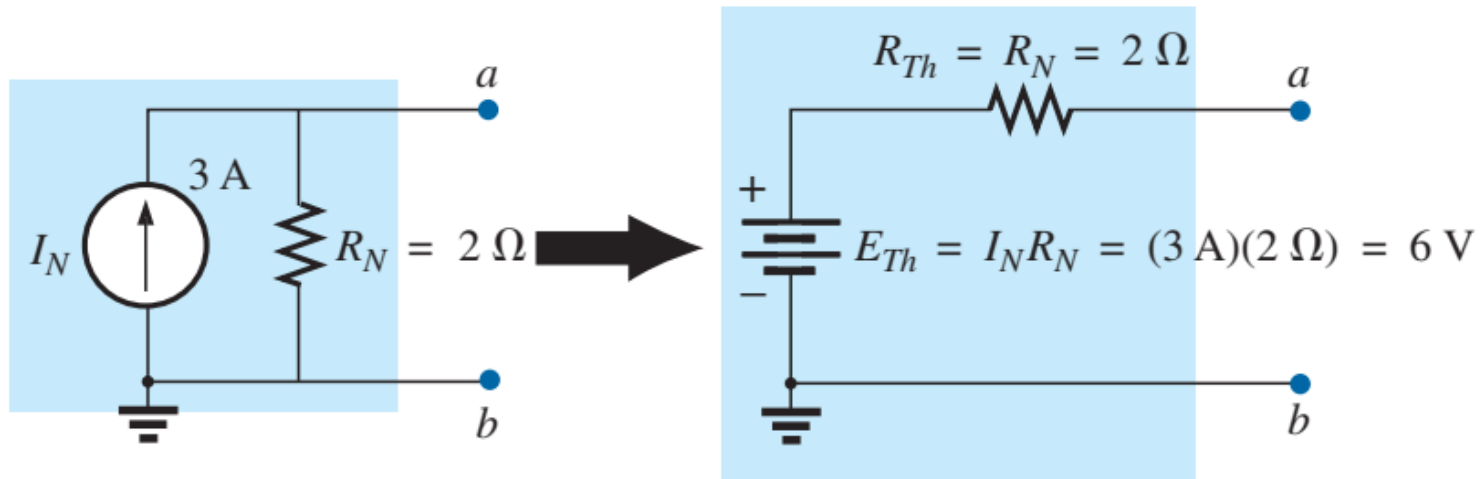
$$I_N = E/R_1$$

3. Norton's Theorem (7/13)

Step 6:



Converting the Norton equivalent circuit in step 6 to a Thévenin equivalent circuit.



3. Norton's Theorem (8/13)

Homework!

EXAMPLE 11 Find the Norton equivalent circuit for the network external to the $9\ \Omega$ resistor in Fig. 22.

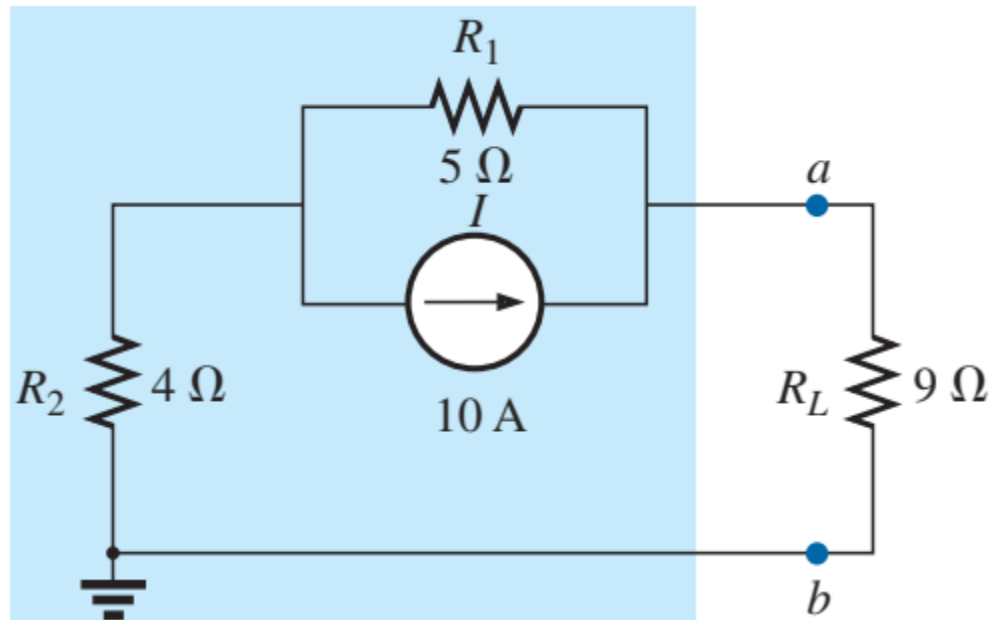


Fig. 22.

3. Norton's Theorem (9/13)

EXAMPLE 12 (Two sources) Find the Norton equivalent circuit for the portion of the network to the left of $a-b$ in Fig. 23.

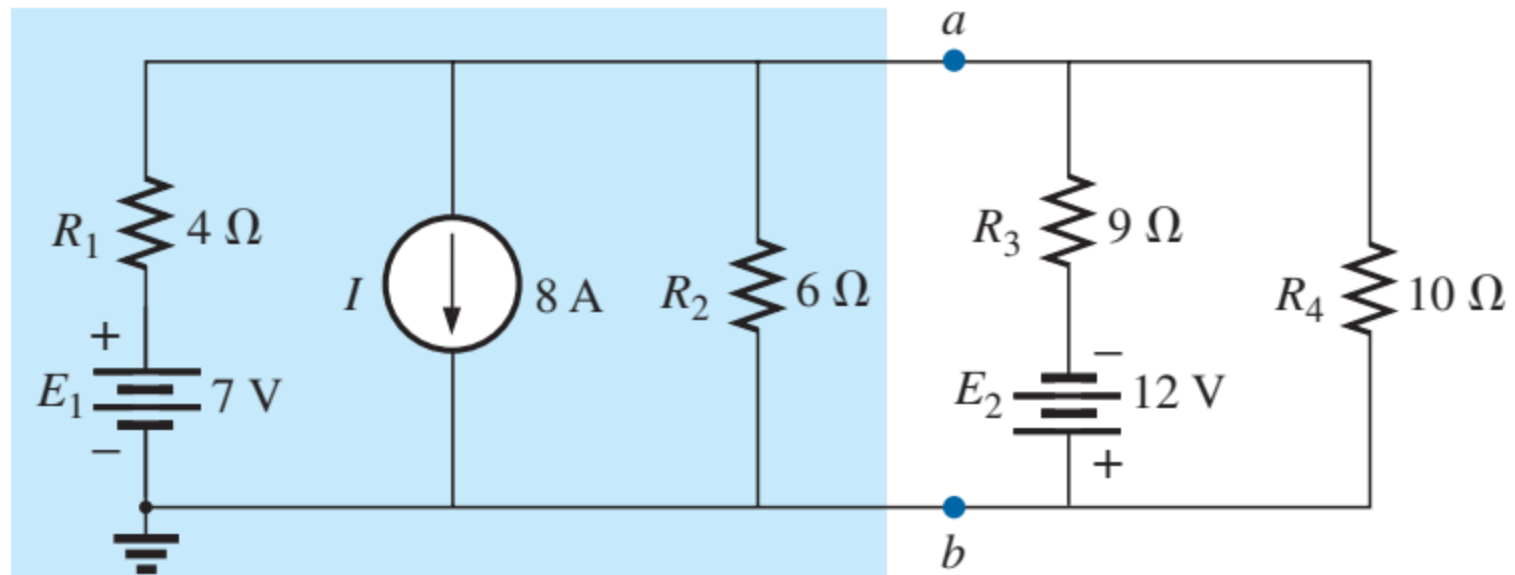
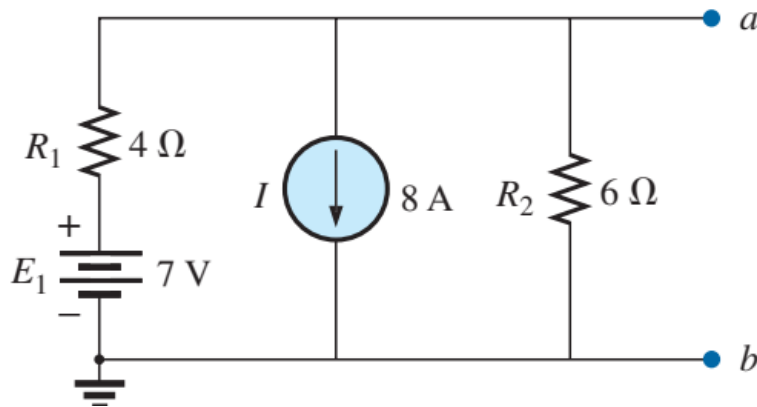


Fig. 23.

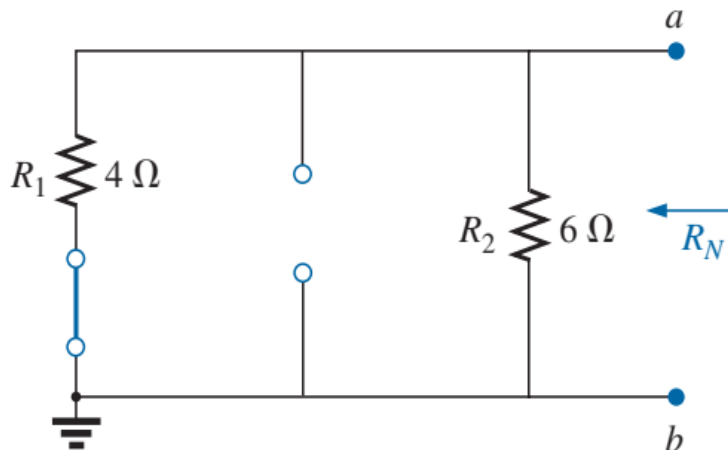
3. Norton's Theorem (10/13)

Solution:

Step 1&2:



Step 3&4:



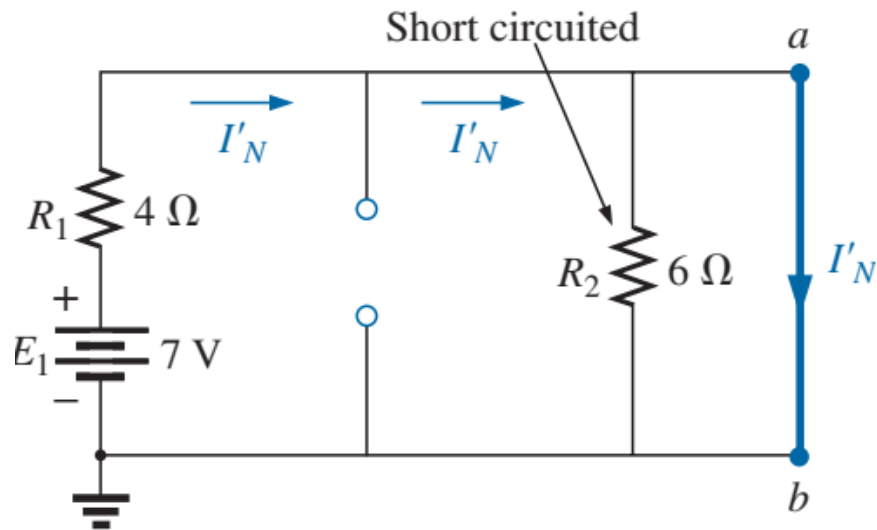
$$R_N = R_1 \parallel R_2 = 4\ \Omega \parallel 6\ \Omega = \frac{(4\ \Omega)(6\ \Omega)}{4\ \Omega + 6\ \Omega} = \frac{24\ \Omega}{10} = 2.4\ \Omega$$

3. Norton's Theorem (11/13)

Solution:

Step 5:

(Using superposition) For the 7 V battery ,



$$I'_N = \frac{E_1}{R_1} = \frac{7\text{ V}}{4\ \Omega} = 1.75\text{ A}$$

3. Norton's Theorem (12/13)

Solution:

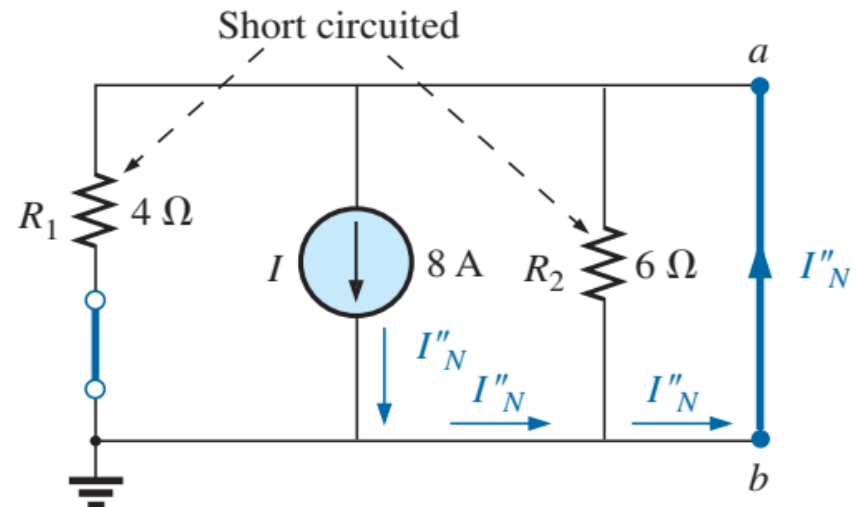
Step 5: (continued)

(Using superposition) For the 8 A source, we find that both R_1 and R_2 have been “short circuited” by the direct connection between a and b.

$$I''_N = I = 8 \text{ A}$$

The result is

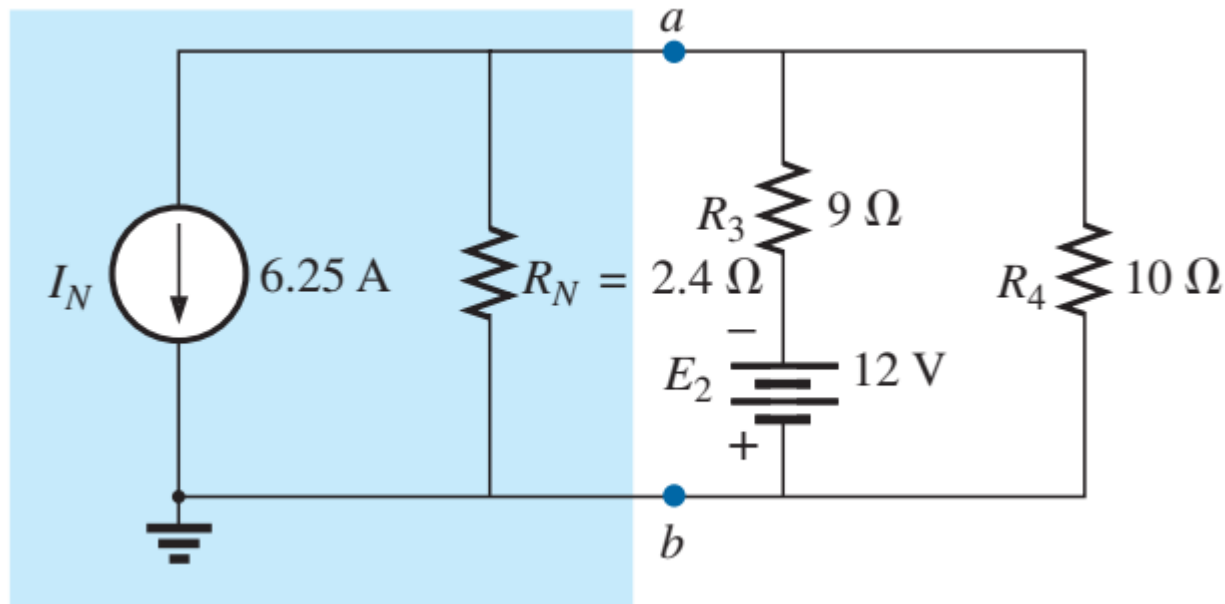
$$I_N = I''_N - I'_N = 8 \text{ A} - 1.75 \text{ A} = \mathbf{6.25 \text{ A}}$$



3. Norton's Theorem (13/13)

Solution:

Step 6:



4. Maximum Power Transfer Theorem (1/17)

- In amplifiers and in most communication circuits such as radio receivers and transmitters, it is often desired that the load receive the maximum amount of power from a source.
- *A load will receive maximum power from a network when its resistance is exactly equal to the Thévenin resistance of the network applied to the load. That is, $R_L = R_{Th} = R_N$.*
- In other words, for the Thévenin equivalent circuit in Fig. 24, when the load is set equal to the Thévenin resistance, the load will receive maximum power from the network.

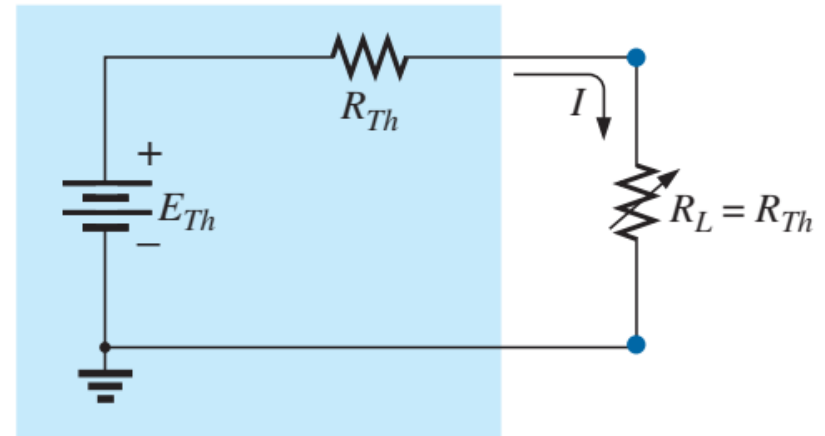


Fig. 24.

4. Maximum Power Transfer Theorem (2/17)

- Using Fig. 24 in previous slide, with $R_L = R_{Th}$, we can determine the maximum power delivered to the load by first finding the current:

$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{E_{Th}}{R_{Th} + R_{Th}} = \frac{E_{Th}}{2R_{Th}}$$

Then we substitute into the power equation:

$$P_L = I_L^2 R_L = \left(\frac{E_{Th}}{2R_{Th}} \right)^2 (R_{Th}) = \frac{E_{Th}^2 R_{Th}}{4R_{Th}^2}$$

and

$$P_{L_{max}} = \frac{E_{Th}^2}{4R_{Th}}$$

Similarly, for Norton's theorem

$$P_{max} = E_{th}^2 / 4 R_N$$

$$E_{th} = I_N R_N$$

$$P_{max} = I_N^2 R_N^2 / 4 R_N$$

$$P_{max} = I_N^2 R_N / 4$$

4. Maximum Power Transfer Theorem (3/17)

- To demonstrate that maximum power is indeed transferred to the load under the conditions defined above, consider the Thévenin equivalent circuit in Fig. 25.
- *Maximum power transfer occurs when the load voltage and current are one-half their maximum possible values.*

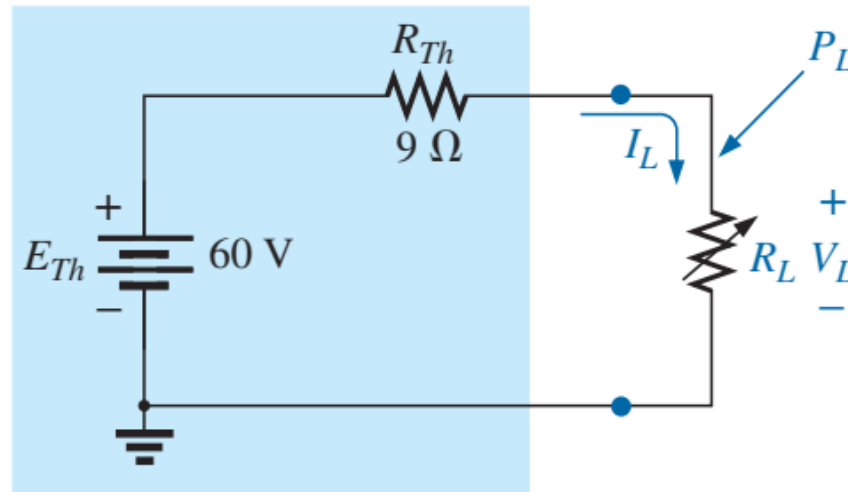


Fig. 25.

4. Maximum Power Transfer Theorem (4/17)

- For the circuit in Fig. 25, the current through the load is determined by

$$I_L = \frac{E_{Th}}{R_{Th} + R_L} = \frac{60 \text{ V}}{9 \Omega + R_L}$$

The voltage is determined by

$$V_L = \frac{R_L E_{Th}}{R_L + R_{Th}} = \frac{R_L (60 \text{ V})}{R_L + R_{Th}}$$

and the power by

$$P_L = I_L^2 R_L = \left(\frac{60 \text{ V}}{9 \Omega + R_L} \right)^2 (R_L) = \frac{3600 R_L}{(9 \Omega + R_L)^2}$$

- If we tabulate the three quantities versus a range of values for R_L from 0.1Ω to 30Ω , we obtain the results appearing in Table 1 in slide 56. Note in particular that when R_L is equal to the Thévenin resistance of 9Ω , the power has a maximum value of 100 W, the current is 3.33 A or one-half its

4. Maximum Power Transfer Theorem (5/17)

- maximum value of 6.60 A (as would result with a short circuit across the output terminals), and the voltage across the load is 30 V, or one-half its maximum value of 60 V (as would result with an open circuit across its output terminals). As you can see, there is no question that maximum power is transferred to the load when the load equals the Thévenin value.

4. Maximum Power Transfer Theorem (6/17)

TABLE 1

$R_L (\Omega)$	$P_L (W)$	$I_L (A)$	$V_L (V)$
0.1	4.35	6.60	0.66
0.2	8.51	6.52	1.30
0.5	19.94	6.32	3.16
1	36.00	6.00	6.00
2	59.50	5.46	10.91
3	75.00	5.00	15.00
4	85.21	4.62	18.46
5	91.84	4.29	21.43
6	96.00	4.00	24.00
7	98.44	3.75	26.25
8	99.65	3.53	28.23
9 (R_{Th})	100.00 (Maximum)	3.33 ($I_{max}/2$)	30.00 ($E_{Th}/2$)
10	99.72	3.16	31.58
11	99.00	3.00	33.00
12	97.96	2.86	34.29
13	96.69	2.73	35.46
14	95.27	2.61	36.52
15	93.75	2.50	37.50
16	92.16	2.40	38.40
17	90.53	2.31	39.23
18	88.89	2.22	40.00
19	87.24	2.14	40.71
20	85.61	2.07	41.38
25	77.86	1.77	44.12
30	71.00	1.54	46.15
40	59.98	1.22	48.98
100	30.30	0.55	55.05
500	6.95	0.12	58.94
1000	3.54	0.06	59.47

4. Maximum Power Transfer Theorem (7/17)

- Under maximum power conditions, only half the power delivered by the source gets to the load. On an efficiency basis, we are working at only a 50% level, but we are content *because we are getting maximum power out of our system*.
- The dc operating efficiency is defined as the ratio of the power delivered to the load (P_L) to the power delivered by the source (P_S). That is,

$$\eta\% = \frac{P_L}{P_s} \times 100\%$$

For the situation where $R_L = R_{Th}$,

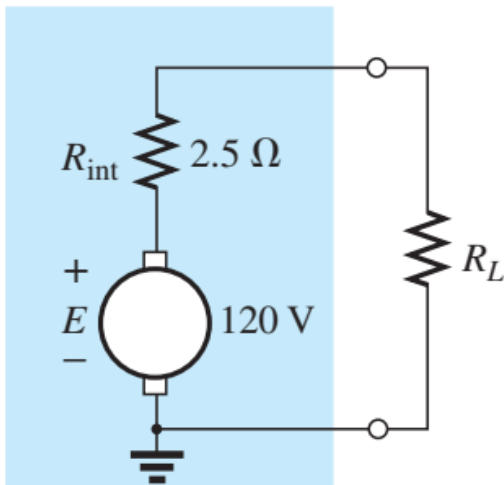
$$\begin{aligned}\eta\% &= \frac{I_L^2 R_L}{I_L^2 R_T} \times 100\% = \frac{R_L}{R_T} \times 100\% = \frac{R_{Th}}{R_{Th} + R_{Th}} \times 100\% \\ &= \frac{R_{Th}}{2R_{Th}} \times 100\% = \frac{1}{2} \times 100\% = \mathbf{50\%}\end{aligned}$$

4. Maximum Power Transfer Theorem (8/17)

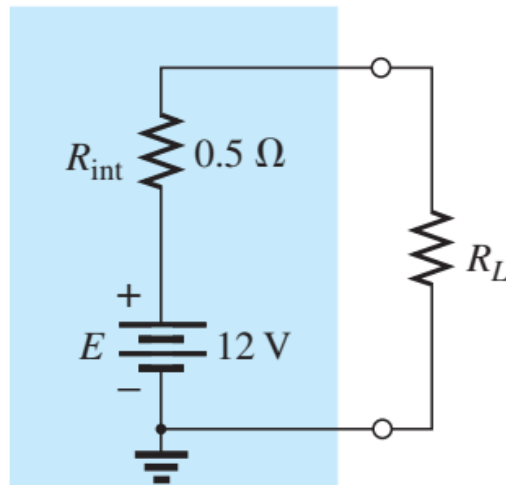
EXAMPLE 13 A dc generator, battery, and laboratory supply are connected to resistive load R_L in Fig. 26 next slide.

- For each, determine the value of R_L for maximum power transfer to R_L .
- Under maximum power conditions, what are the current level and the power to the load for each configuration?
- What is the efficiency of operation for each supply in part (b)?
- If a load of $1\text{ k}\Omega$ were applied to the laboratory supply, what would the power delivered to the load be? Compare your answer to the level of part (b). What is the level of efficiency?
- For each supply, determine the value of R_L for 75% efficiency.

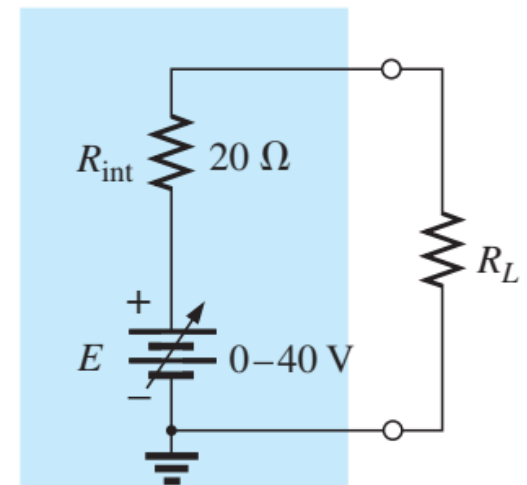
4. Maximum Power Transfer Theorem (9/17)



(a) dc generator



(b) Battery



(c) Laboratory supply

Fig. 26.

4. Maximum Power Transfer Theorem (10/17)

Solutions:

a. For the dc generator,

$$R_L = R_{Th} = R_{int} = \mathbf{2.5\ \Omega}$$

For the 12 V car battery,

$$R_L = R_{Th} = R_{int} = \mathbf{0.05\ \Omega}$$

For the dc laboratory supply,

$$R_L = R_{Th} = R_{int} = \mathbf{20\ \Omega}$$

b. For the dc generator,

$$P_{L_{\max}} = \frac{E_{Th}^2}{4R_{Th}} = \frac{E^2}{4R_{int}} = \frac{(120\text{ V})^2}{4(2.5\ \Omega)} = \mathbf{1.44\text{ kW}}$$

For the 12 V car battery,

$$P_{L_{\max}} = \frac{E_{Th}^2}{4R_{Th}} = \frac{E^2}{4R_{int}} = \frac{(12\text{ V})^2}{4(0.05\ \Omega)} = \mathbf{720\text{ W}}$$

4. Maximum Power Transfer Theorem (11/17)

For the dc laboratory supply,

$$P_{L_{\max}} = \frac{E_{Th}^2}{4R_{Th}} = \frac{E^2}{4R_{\text{int}}} = \frac{(40 \text{ V})^2}{4(20 \Omega)} = \mathbf{20 \text{ W}}$$

- c. They are all operating under a 50% efficiency level because $R_L = R_{Th}$.
- d. The power to the load is determined as follows:

$$I_L = \frac{E}{R_{\text{int}} + R_L} = \frac{40 \text{ V}}{20 \Omega + 1000 \Omega} = \frac{40 \text{ V}}{1020 \Omega} = 39.22 \text{ mA}$$

and $P_L = I_L^2 R_L = (39.22 \text{ mA})^2 (1000 \Omega) = \mathbf{1.54 \text{ W}}$

The power level is significantly less than the 20 W achieved in part (b). The efficiency level is

$$\begin{aligned} \eta\% &= \frac{P_L}{P_s} \times 100\% = \frac{1.54 \text{ W}}{EI_s} \times 100\% = \frac{1.54 \text{ W}}{(40 \text{ V})(39.22 \text{ mA})} \times 100\% \\ &= \frac{1.54 \text{ W}}{1.57 \text{ W}} \times 100\% = \mathbf{98.09\%} \end{aligned}$$

which is markedly higher than achieved under maximum power conditions

4. Maximum Power Transfer Theorem (13/17)

e. For the dc generator,

$$\eta = \frac{P_o}{P_s} = \frac{R_L}{R_{Th} + R_L} \text{ (\eta in decimal form)}$$

and

$$\eta = \frac{R_L}{R_{Th} + R_L}$$

$$\eta(R_{Th} + R_L) = R_L$$

$$\eta R_{Th} + \eta R_L = R_L$$

$$R_L(1 - \eta) = \eta R_{Th}$$

and

$$R_L = \frac{\eta R_{Th}}{1 - \eta}$$

$$R_L = \frac{0.75(2.5 \, \Omega)}{1 - 0.75} = \mathbf{7.5 \, \Omega}$$

For the battery,

$$R_L = \frac{0.75(0.05 \, \Omega)}{1 - 0.75} = \mathbf{0.15 \, \Omega}$$

4. Maximum Power Transfer Theorem (14/17)

For the laboratory supply,

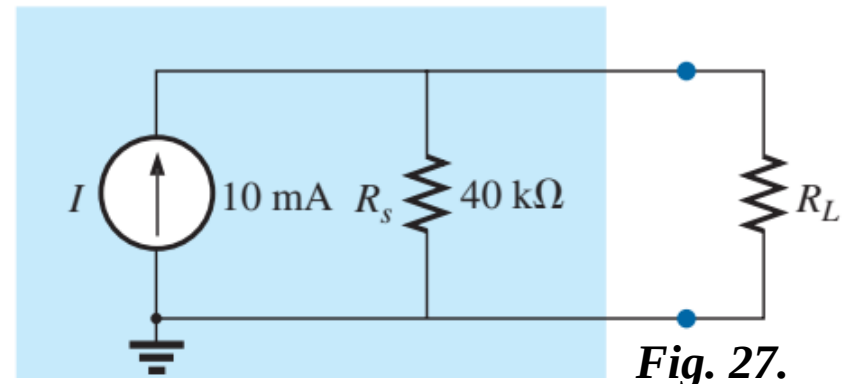
$$R_L = \frac{0.75(20\ \Omega)}{1 - 0.75} = \mathbf{60\ \Omega}$$

4. Maximum Power Transfer Theorem (15/17)

Homework!

EXAMPLE 14 The analysis of a transistor network resulted in the reduced equivalent in Fig. 27.

- Find the load resistance that will result in maximum power transfer to the load, and find the maximum power delivered.
- If the load were changed to $68\text{ k}\Omega$, would you expect a fairly high level of power transfer to the load based on the results of part (a)? What would the new power level be? Is your initial assumption verified?
- If the load were changed to $8.2\text{ k}\Omega$, would you expect a fairly high level of power transfer to the load based on the results of part (a)? What would the new power level be? Is your initial assumption verified?



4. Maximum Power Transfer Theorem (16/17)

Homework!

EXAMPLE 15 In Fig. 28.a fixed load of $16\ \Omega$ is applied to a 48 V supply with an internal resistance of $36\ \Omega$.

- For the conditions in Fig.28. what is the power delivered to the load and lost to the internal resistance of the supply?
- If the designer has some control over the internal resistance level of the supply, what value should he or she make it for maximum power to the load? What is the maximum power to the load? How does it compare to the level obtained in part (a)?
- Without making a single calculation, find the value that would result in more power to the load if the designer could change the internal resistance to $22\ \Omega$ or $8.2\ \Omega$. Verify your conclusion by calculating the power to the load for each value.

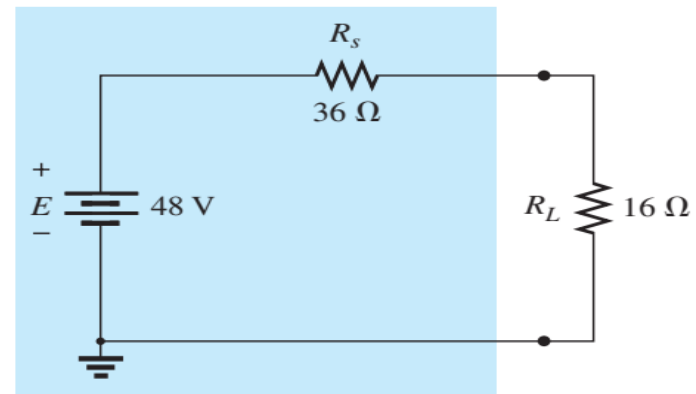


Fig. 28.

4. Maximum Power Transfer Theorem (17/17)

Homework!

EXAMPLE 16 Given the network in Fig.29. find the value of R_L for maximum power to the load, and find the maximum power to the load.

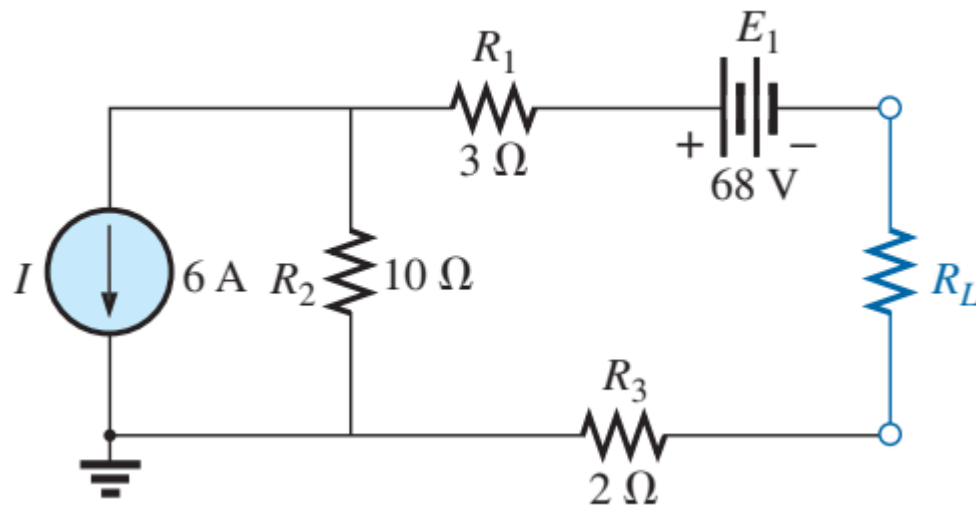


Fig. 29.

References: Textbooks

1. *Robert L.Boylestad. (2014). Introductory Circuit Analysis. 12th Edition. Pearson Education Limited.*
2. *Allan H.Robbins, Wilhelm C.Miller. (2012). Circuit Analysis Theory and Practice. 5th Edition. Cengage Learning.*
3. *Charles K.Alexander, Matthew N.O.Sadiku (2017). Fundamentals of Electric Circuits. 6th Edition. McGraw Hill Education.*